

Uncertainty quantification for stationary and time-dependent PDEs subject to Gevrey regular random domain deformations

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We study uncertainty quantification for partial differential equations subject to domain uncertainty. We parameterize the random domain using the model recently considered by Chernov and Lê (2024) as well as Harbrecht, Schmidlin, and Schwab (2024) in which the input random field is assumed to belong to a Gevrey smoothness class. This approach has the advantage of being substantially more general than models which assume a particular parametric representation of the input random field such as a Karhunen–Loève series expansion. We consider both the Poisson equation as well as the heat equation and design randomly shifted lattice quasi-Monte Carlo (QMC) cubature rules for the computation of the expected solution under domain uncertainty. We show that these QMC rules exhibit dimension-independent, essentially linear cubature convergence rates in this framework. In addition, we complete the error analysis by taking into account the approximation errors incurred by dimension truncation of the random input field and finite element discretization. Numerical experiments are presented to confirm the theoretical rates.

Keywords: Quasi-Monte Carlo method; Random domain; Uncertainty quantification; Partial differential equation; Gevrey regularity.

1. Introduction

Domain uncertainties arise frequently in numerous applications due to factors such as restricted resolution and noise in measurements or manufacturing tolerances. In particular, various biological processes are naturally posed on uncertain domains. Inaccuracies in surface imaging such as microscopy, tomography, and in product manufacturing lead to the problem of describing a variety of different phenomena on randomly deformed domains, see, e.g., [Corrado et al. \(2023\)](#) and [Tekkaya et al. \(2017\)](#). In recent years, there has been increasing attention on the topic ([Castrillón-Candás et al., 2016](#); [Church et al., 2020](#); [Djurdevac, 2021](#); [Dölz et al., 2022](#); [Harbrecht et al., 2016](#); [Hiptmair et al., 2018](#); [Xiu and Tartakovsky, 2006](#)), particularly from a numerical perspective.

The natural first question that arises is how to represent an uncertain domain and the quantities of interest to be considered. A common way of representing the random domain is to assume the existence of a random field $\mathbf{V}(\omega)$ subject to a probability space $(\Omega, \mathcal{A}, \mathbb{P})$ that maps a given deterministic domain $D_{\text{ref}} \subset \mathbb{R}^d$, $d \in \{1, 2, 3\}$, to a random domain $D(\omega)$ for $\omega \in \Omega$. This is the basis of the so-called *domain mapping method* ([Harbrecht et al., 2016](#); [Xiu and Tartakovsky, 2006](#)) which we will also adopt in the

present study: if u denotes the solution to a partial differential equation (PDE) on a random domain $D(\omega)$, then the pullback solution is defined as the composition $\hat{u} = u \circ \mathbf{V}(\omega)$. If the domain mapping $\mathbf{V}(\omega)$ is sufficiently smooth with respect to the spatial variable (e.g., it is a $\mathcal{C}^2$ -diffeomorphism), then the pullback solution $\hat{u}$ can be expressed as the solution to a transformed PDE problem posed on the reference domain, equipped with a random coefficient and a random source term taking into account the effect of the random deformation. This reformulation allows us to deal with potentially large deformations and to apply numerous available numerical methods for solving equations with random coefficients on a fixed domain.

A common alternative to the domain mapping method is the so-called perturbation method (Harbrecht et al., 2008), where one starts with a perturbation field defined on the boundary of a reference domain and a shape Taylor expansion with respect to this field is used to express the solution of the considered equation. The main disadvantage of this method is that it is applicable only for small perturbations. In contrast to the domain mapping method, one only needs to define the random transformation on the boundary ∂D_{ref} instead of the closure of the entire domain $\overline{D_{\text{ref}}}$. However, the perturbation approach requires both the reference domain and random perturbations to be $\mathcal{C}^2$ -smooth, while the domain mapping method allows the reference domain D_{ref} to be Lipschitz with only the random perturbations $\mathbf{V}(\omega)$ required to be $\mathcal{C}^2$ -smooth.

For the computation of the expected solution, we will use lattice quasi-Monte Carlo (QMC) cubature rules. QMC methods are a popular method for assessing the response statistics of PDEs with random inputs, exhibiting dimension-independent, faster-than-Monte Carlo cubature convergence rates under moderate assumptions placed on the input random field (Graham et al., 2015, 2011; Herrmann and Schwab, 2019; Kuo and Nuyens, 2016a, 2018; Kuo et al., 2012).

In a recent pair of articles Chernov and Lê (2024a,b) demonstrated that the input random field can be generalized beyond those considered in the previously existing literature while still retaining dimension-independent QMC convergence rates for the PDE response. They showed that it is enough to assume that the input random field satisfies a certain parametric regularity bound which ensures that the PDE response belong to the so-called Gevrey smoothness class. For details on Gevrey class functions and their properties we refer to Rodino (1993). Moreover, Harbrecht et al. (2024) showed that Gevrey regularity of parametric inputs is inherited by the solutions to a very general class of operator equations. We remark that some of the parametric regularity analysis carried out in Harbrecht et al. (2024) may be applied to the stationary PDE problem considered in our work. However, since the focus in Harbrecht et al. (2024) is not the application of numerical methods, certain constants appearing in their parametric regularity estimates are not tracked explicitly. In contrast, since our goal is the design of constructible QMC cubatures for the computation of response statistics, it is necessary to make the dependence of the parametric regularity bounds on the model parameters explicit in our analysis. In addition, we consider the parabolic case which is not treated in Harbrecht et al. (2024). In general, domain uncertainty quantification has focused on stationary problems in the existing literature. Nevertheless, there has been recent interest on quantifying domain uncertainty for parabolic PDEs (Castrillón-Candás and Xu, 2021; Djurdjevac, 2021).

Under the assumption of Gevrey regularity imposed on the random domain mapping, we can prove that the solutions to both, the Poisson equation and the heat equation on a random domain, belong to a Gevrey class. After truncating the infinite-dimensional input vector field $\mathbf{V}$ to s dimensions, we can develop tailored QMC point sets achieving essentially linear cubature convergence rates independently of the stochastic dimension s of the problem. The cubature convergence rates will be assessed in a series of numerical experiments. In addition, we shall also briefly address the other sources of discretization error stemming from dimension truncation of the input random field as well as finite

element approximation of the PDE solution. Together they yield an overall error estimate with explicit decay rates with respect to the truncation dimension, spatial and temporal refinement levels, and the QMC convergence rate.

Contributions. This paper presents the following key contributions:

- The Gevrey class accommodates very general parameterizations of domain uncertainty. Moreover, in addition to the Poisson equation, we also study domain uncertainty quantification for a *parabolic* PDE.
- We develop a novel parametric regularity analysis for these problems, which is used to design tailored *QMC cubature rules* with essentially linear convergence rates *independently of the truncation dimension*.
- We consider the approximation errors incurred by dimension truncation of the random domain mapping and finite element discretization.
- Numerical experiments are presented which confirm the sharpness of our results.

Outline of the paper. This paper is organized as follows. The elliptic and parabolic PDE problems as well as the Gevrey regular parameterization of domain uncertainty are introduced in Section 2 alongside the definitions and main assumptions required for the subsequent analysis. In Section 3, we briefly review the basic properties of randomly shifted rank-1 lattice rules. The parametric regularity analysis for the considered model problems is carried out in Section 4. The corresponding error rates for dimension truncation, finite element discretization, and QMC integration are stated in Section 5. Numerical experiments are presented to assess the QMC convergence rate for both the stationary and time-dependent cases in Section 6. Some conclusions on our findings are drawn in Section 7. The appendix covers results on technical recurrence relations and PDE regularity used in our analysis.

2. Notations, preliminaries, and assumptions

Let $U := [-\frac{1}{2}, \frac{1}{2}]^{\mathbb{N}}$ be the set of parameters. We define the set of finitely-supported multi-indices by

$$\mathcal{F} := \left\{ \mathbf{v} \in \mathbb{N}_0^{\mathbb{N}} \mid |\mathbf{v}| := \sum_{j \geq 1} v_j < \infty \right\}.$$

Let $\mathbf{m}, \mathbf{v} \in \mathcal{F}$ be multi-indices and let $\mathbf{b} = (b_j)_{j \geq 1}$ be a sequence of real numbers. The support of a multi-index $\mathbf{v}$ is defined as

$$\text{supp}(\mathbf{v}) := \{j \in \mathbb{N} \mid v_j \neq 0\}.$$

In what follows, we shall use the following multi-index notations:

$$\mathbf{m} \leq \mathbf{v} \quad \Leftrightarrow \quad m_j \leq v_j \quad \text{for all } j \in \mathbb{N},$$

$$\binom{\mathbf{v}}{\mathbf{m}} = \prod_{j \geq 1} \binom{v_j}{m_j}, \quad \mathbf{b}^{\mathbf{v}} = \prod_{j \in \text{supp}(\mathbf{v})} b_j^{v_j}, \quad \partial_{\mathbf{y}}^{\mathbf{v}} = \prod_{j \in \text{supp}(\mathbf{v})} \frac{\partial^{v_j}}{\partial y_j^{v_j}},$$

where $\mathbf{y} \in U$ and we use the convention $0^0 := 1$.

Let $D \subset \mathbb{R}^d$ be a non-empty Lipschitz domain for $d \in \{1, 2, 3\}$. We denote the subspace of $H^1(D)$ with vanishing trace on ∂D as $H_0^1(D)$. The dual space of $H_0^1(D)$ is denoted by $H^{-1}(D)$ with respect to the pivot space $L^2(D)$.

Definition 1. Let $D \subset \mathbb{R}^d$ be a non-empty Lipschitz domain for $d \in \{1, 2, 3\}$. We define

$$\|v\|_{L^\infty(D)} := \begin{cases} \operatorname{ess\,sup}_{\mathbf{x} \in D} \|v(\mathbf{x})\|_{\ell^2} & \text{if } v: D \rightarrow \mathbb{R}^d, \\ \operatorname{ess\,sup}_{\mathbf{x} \in D} \|v(\mathbf{x})\|_2 & \text{if } v: D \rightarrow \mathbb{R}^{d \times d}, \end{cases}$$

where $\|\cdot\|_{\ell^2}$ denotes the Euclidean norm and $\|\cdot\|_2$ denotes the matrix spectral norm. In addition, if $v: D \rightarrow \mathbb{R}^d$, we define

$$\|v\|_{W^{1,\infty}(D)} := \max \left\{ \operatorname{ess\,sup}_{\mathbf{x} \in D} \|v(\mathbf{x})\|_{\ell^2}, \operatorname{ess\,sup}_{\mathbf{x} \in D} \|v'(\mathbf{x})\|_2 \right\},$$

where $v': D \rightarrow \mathbb{R}^{d \times d}$ denotes the Jacobian matrix of v . Furthermore, we will make use of the Sobolev norms defined as

$$\begin{aligned} \|v\|_{H_0^1(D)} &:= \|\nabla v\|_{L^2(D)} = \left(\int_D \|\nabla v(\mathbf{x})\|_{\ell^2}^2 d\mathbf{x} \right)^{\frac{1}{2}}, \quad v \in H_0^1(D), \\ \|v\|_{H_0^1(D) \cap H^2(D)} &:= \left(\|v\|_{L^2(D)}^2 + \|\Delta v\|_{L^2(D)}^2 \right)^{\frac{1}{2}}, \quad v \in H_0^1(D) \cap H^2(D). \end{aligned}$$

Furthermore, when $v \in \mathcal{C}^k(\overline{D})$ for $k \in \mathbb{N}_0$, we define

$$\|v\|_{\mathcal{C}^k(\overline{D})} := \max_{|\mathbf{v}| \leq k} \sup_{\mathbf{x} \in \overline{D}} |\partial_{\mathbf{x}}^{\mathbf{v}} v(\mathbf{x})|.$$

We omit the domain D from the subscript when there is no risk of confusion.

Furthermore, if V is a Sobolev space and $I \subset \mathbb{R}$ is an interval, we denote by $L^2(I; V)$ the space of all measurable functions $v: I \rightarrow V$ with finite norm

$$\|v\|_{L^2(I; V)} := \left(\int_I \|v(\cdot, t)\|_V^2 dt \right)^{1/2}.$$

We assume that the reference domain $D_{\text{ref}} \subset \mathbb{R}^d$, $d \in \{1, 2, 3\}$, is bounded and non-empty with Lipschitz regular boundary, and that the mapping

$$\mathbf{V}: \overline{D_{\text{ref}}} \times U \rightarrow \mathbb{R}^d$$

is a vector field. We define the family of admissible domains $\{D(\mathbf{y})\}_{\mathbf{y} \in U}$, parameterized by

$$D(\mathbf{y}) := \mathbf{V}(D_{\text{ref}}, \mathbf{y}), \quad \mathbf{y} \in U,$$

and the *hold-all domain* is defined by setting

$$\mathcal{D} := \bigcup_{\mathbf{y} \in U} D(\mathbf{y}). \tag{2.1}$$

Furthermore, we denote by $J(\cdot, \mathbf{y}): D_{\text{ref}} \rightarrow \mathbb{R}^{d \times d}$ the Jacobian matrix of vector field $\mathbf{V}$ with respect to $\mathbf{x} \in D_{\text{ref}}$. In addition, we define the dimensionally truncated domain mapping $\mathbf{V}_s: \overline{D_{\text{ref}}} \times U \rightarrow \mathbb{R}^d$ by setting

$$\mathbf{V}_s(\mathbf{x}, \mathbf{y}) := \mathbf{V}(\mathbf{x}, (y_1, \dots, y_s, 0, 0, \dots)), \quad \mathbf{x} \in \overline{D_{\text{ref}}} \times U.$$

Instead of following the approaches taken by [Harbrecht et al. \(2016\)](#) or [Hakula et al. \(2024\)](#), we carry out the analysis using the arguments of [Chernov and Lê \(2024a,b\)](#). As noted by [Harbrecht and Schmidlin](#)

(2022, Remark 1), instead of assuming a particular kind of *parametric representation* for $\mathbf{V}(\mathbf{x}, \mathbf{y})$, such as a Karhunen–Loève expansion, we can instead only impose a condition on the growth rate of the parametric derivatives of $\mathbf{V}(\mathbf{x}, \mathbf{y})$. We assume that the random domain mapping $\mathbf{V}$ is parameterized by a countable sequence $\mathbf{y} = (y_j)_{j \geq 1}$ of independent and identically distributed (i.i.d.) uniform random variables $y_j = y_j(\omega)$, $j \geq 1$, supported on $[-\frac{1}{2}, \frac{1}{2}]$. The main modeling assumptions regarding the random domain mapping and other problem parameters are stated below. Crucially for our analysis, the parametric regularity bound for the input random field is quantified in Assumption (A3).

Assumptions.

(A1) We assume that for each $\mathbf{y} \in U$, $\mathbf{V}(\cdot, \mathbf{y}) : \overline{D_{\text{ref}}} \rightarrow \mathbb{R}^d$ is an invertible and twice continuously differentiable vector field, with

$$\sup_{\mathbf{y} \in U} \|\mathbf{V}(\cdot, \mathbf{y})\|_{\mathcal{C}^2(\overline{D_{\text{ref}}})} < \infty \quad \text{and} \quad \sup_{\mathbf{y} \in U} \|\mathbf{V}^{-1}(\cdot, \mathbf{y})\|_{\mathcal{C}^2(\overline{D(\mathbf{y})})} < \infty.$$

(A2) We assume that there exist uniform bounds $0 < \sigma_{\min} \leq 1 \leq \sigma_{\max} < \infty$ such that for all $\mathbf{x} \in D_{\text{ref}}$, $\mathbf{y} \in U$

$$\sigma_{\min} \leq \min \sigma(J(\mathbf{x}, \mathbf{y})) \leq \max \sigma(J(\mathbf{x}, \mathbf{y})) \leq \sigma_{\max},$$

holds, where $\sigma(M)$ denotes the set of all singular values of matrix M .

(A3) **Gevrey regularity of the domain mapping:** We assume that the vector field $\mathbf{V}(\cdot, \mathbf{y})$ is infinitely many times continuously differentiable with respect to $\mathbf{y} \in U$ and there exists a constant $C \geq 1$, an exponent $\beta \geq 1$, and a sequence $\mathbf{b} = (b_j)_{j \geq 1}$ of non-negative numbers such that

$$\|\partial_{\mathbf{y}}^{\mathbf{v}} \mathbf{V}(\cdot, \mathbf{y})\|_{W^{1,\infty}(D_{\text{ref}})} \leq C(|\mathbf{v}|!)^{\beta} \mathbf{b}^{\mathbf{v}} \quad \text{for all } \mathbf{v} \in \mathcal{F}, \mathbf{y} \in U.$$

It is an immediate consequence of the above condition that

$$\|\partial_{\mathbf{y}}^{\mathbf{v}} J(\cdot, \mathbf{y})\|_{L^{\infty}(D_{\text{ref}})} \leq C(|\mathbf{v}|!)^{\beta} \mathbf{b}^{\mathbf{v}} \quad \text{for all } \mathbf{v} \in \mathcal{F}, \mathbf{y} \in U.$$

Furthermore, it follows by continuity that

$$\lim_{s \rightarrow \infty} \mathbf{V}(\cdot, \mathbf{y}_{\leq s}) = \mathbf{V}(\cdot, \lim_{s \rightarrow \infty} \mathbf{y}_{\leq s}) = \mathbf{V}(\cdot, \mathbf{y})$$

and hence, for all $\mathbf{y} \in U$ there holds

$$\lim_{s \rightarrow \infty} \|\mathbf{V}(\cdot, \mathbf{y}) - \mathbf{V}_s(\cdot, \mathbf{y})\|_{W^{1,\infty}(D_{\text{ref}})} = 0.$$

(A4) **Gevrey regularity of the source term:** The source term $f : \mathcal{D} \rightarrow \mathbb{R}$ is infinitely many times continuously differentiable and satisfies

$$\|\partial_{\mathbf{x}}^{\mathbf{v}} f\|_{L^{\infty}(\mathcal{D})} \leq C_f(|\mathbf{v}|!)^{\beta} \mathbf{p}^{\mathbf{v}} \quad \text{for all } \mathbf{v} \in \mathcal{F},$$

where $C_f > 0$ is a constant depending on f and $\mathbf{p} = (\rho_j)_{j=1}^d$ is a sequence of non-negative real numbers. Without loss of generality, we assume that the Gevrey exponent β coincides with that of Assumption (A3).

(A5) The sequence $\mathbf{b}$ in Assumption (A3) satisfies $\mathbf{b} \in \ell^p(\mathbb{N})$ for some $p \in (0, 1)$.

(A6) The sequence $\mathbf{b}$ in Assumption (A3) satisfies

$$b_1 \geq b_2 \geq b_3 \geq \dots.$$

(A7) The reference domain $D_{\text{ref}} \subset \mathbb{R}^d$ is a convex polyhedron.

We remark that assumption (A7) can be relaxed if one allows for curvilinear triangulations or non domain-conformity. However, for simplicity we only use finite elements on domain conforming non-curvilinear triangulations.

In this article we shall consider two model problems subject to domain uncertainty.

Poisson equation. The Poisson equation over a parametric domain $D(\mathbf{y})$ is the problem of finding, for all $\mathbf{y} \in U$, $u(\cdot, \mathbf{y}) : D(\mathbf{y}) \rightarrow \mathbb{R}$ such that

$$\begin{cases} -\Delta u(\mathbf{x}, \mathbf{y}) = f(\mathbf{x}) & \text{for } \mathbf{x} \in D(\mathbf{y}), \\ u(\mathbf{x}, \mathbf{y}) = 0 & \text{for } \mathbf{x} \in \partial D(\mathbf{y}), \end{cases}$$

where $f : \mathcal{D} \rightarrow \mathbb{R}$ is a given source term over the hold-all domain $\mathcal{D}$, see (2.1). The weak formulation over $D(\mathbf{y})$ can be stated as: for each $\mathbf{y} \in U$, find $u(\cdot, \mathbf{y}) \in H_0^1(D(\mathbf{y}))$ such that

$$\int_{D(\mathbf{y})} \nabla u(\mathbf{x}, \mathbf{y}) \cdot \nabla v(\mathbf{x}) \, d\mathbf{x} = \int_{D(\mathbf{y})} f(\mathbf{x}) v(\mathbf{x}) \, d\mathbf{x} \quad \text{for all } v \in H_0^1(D(\mathbf{y})).$$

We consider the path-wise formulation since this allows for straightforward analysis of the higher-order partial derivatives of the PDE solution with respect to the parameter $\mathbf{y}$. However, for the purposes of analysis, it is convenient to analyze instead the pullback solution $\hat{u}(\mathbf{x}, \mathbf{y}) = u(\mathbf{V}(\mathbf{x}, \mathbf{y}), \mathbf{y})$. This is equivalent to the following pullback weak formulation obtained using a change of variables: for all $\mathbf{y} \in U$, find $\hat{u}(\cdot, \mathbf{y}) \in H_0^1(D_{\text{ref}})$ such that

$$\int_{D_{\text{ref}}} (A(\mathbf{x}, \mathbf{y}) \nabla \hat{u}(\mathbf{x}, \mathbf{y})) \cdot \nabla \hat{v}(\mathbf{x}) \, d\mathbf{x} = \int_{D_{\text{ref}}} f_{\text{ref}}(\mathbf{x}, \mathbf{y}) \hat{v}(\mathbf{x}) \, d\mathbf{x} \quad (2.2)$$

for all $\hat{v} \in H_0^1(D_{\text{ref}})$, where

$$\begin{aligned} f_{\text{ref}}(\mathbf{x}, \mathbf{y}) &= f(\mathbf{V}(\mathbf{x}, \mathbf{y})) \det J(\mathbf{x}, \mathbf{y}) \quad \text{and} \\ A(\mathbf{x}, \mathbf{y}) &= (J(\mathbf{x}, \mathbf{y})^T J(\mathbf{x}, \mathbf{y}))^{-1} \det J(\mathbf{x}, \mathbf{y}). \end{aligned}$$

Note that the test functions $\hat{v}$ are independent of the parameter $\mathbf{y} \in U$ due to the fact that the spaces $H_0^1(D_{\text{ref}})$ and $H_0^1(D(\mathbf{y}))$ are isomorphic as shown by Harbrecht et al. (2016, Lemma 1). Since the parameters $\mathbf{y} \in U$ are uniformly bounded, it is a consequence of the Lax–Milgram lemma that $\hat{u}$ can be bounded independently of $\mathbf{y}$. In particular, $\hat{u} \in L^p(U; H_0^1(D_{\text{ref}}))$ for all $1 \leq p \leq \infty$.

Heat equation. The heat equation over a parametric domain $D(\mathbf{y})$ and the time interval $I = [0, T]$ for some fixed terminal time $T > 0$ is the problem of finding, for all $\mathbf{y} \in U$, $u(\cdot, \cdot, \mathbf{y}) : D(\mathbf{y}) \times I \rightarrow \mathbb{R}$ such that

$$\begin{cases} \frac{\partial}{\partial t} u(\mathbf{x}, t, \mathbf{y}) - \Delta u(\mathbf{x}, t, \mathbf{y}) = f(\mathbf{x}) & \text{for } \mathbf{x} \in D(\mathbf{y}), t \in (0, T] \\ u(\mathbf{x}, 0, \mathbf{y}) = u_0(\mathbf{x}), & \text{for } \mathbf{x} \in D(\mathbf{y}), \\ u(\mathbf{x}, t, \mathbf{y}) = 0 & \text{for } \mathbf{x} \in \partial D(\mathbf{y}), t \in (0, T], \end{cases}$$

where $f : \mathcal{D} \rightarrow \mathbb{R}$ is a given source term and $u_0 : \mathcal{D} \rightarrow \mathbb{R}$ a given initial condition. We shall consider the space-time weak formulation of this model problem (cf., e.g., Kunoth and Schwab (2013); Schwab

and Stevenson (2009)), since this is independent of the chosen PDE discretization scheme, thereby accommodating a very general analysis of the parametric regularity of the PDE solution. Moreover, it allows us to conveniently enforce the initial condition almost everywhere as part of the variational formulation. The space-time weak formulation over $D(\mathbf{y}) \times I$ can be stated as:

$$\begin{aligned} & \int_I \int_{D(\mathbf{y})} \frac{\partial}{\partial t} u(\mathbf{x}, t, \mathbf{y}) v_1(\mathbf{x}, t) \, d\mathbf{x} \, dt + \int_I \int_{D(\mathbf{y})} \nabla u(\mathbf{x}, t, \mathbf{y}) \cdot \nabla v_1(\mathbf{x}, t) \, d\mathbf{x} \, dt \\ & + \int_{D(\mathbf{y})} u(\mathbf{x}, 0, \mathbf{y}) v_2(\mathbf{x}) \, d\mathbf{x} \\ & = \int_I \int_{D(\mathbf{y})} f(\mathbf{x}) v_1(\mathbf{x}, t) \, d\mathbf{x} \, dt + \int_I \int_{D(\mathbf{y})} u_0(\mathbf{x}) v_2(\mathbf{x}) \, d\mathbf{x} \end{aligned}$$

for all $(v_1, v_2) \in L^2(I; H_0^1(D(\mathbf{y}))) \times L^2(D(\mathbf{y}))$. However, for the purposes of analysis, it is convenient to analyze instead the pullback solution given by

$$\hat{u}(\mathbf{x}, t, \mathbf{y}) = u(\mathbf{V}(\mathbf{x}, \mathbf{y}), t, \mathbf{y}).$$

By change of variables, the pullback space-time variational formulation is given by

$$b(\mathbf{y}; \hat{u}, \hat{v}) = F(\mathbf{y}; \hat{v}) \quad \text{for all } \hat{v} = (\hat{v}_1, \hat{v}_2) \in \mathcal{V} \text{ and } \mathbf{y} \in U, \quad (2.3)$$

where

$$\begin{aligned} b(\mathbf{y}; \hat{u}, \hat{v}) &:= \int_I \int_{D_{\text{ref}}} \frac{\partial}{\partial t} \hat{u}(\mathbf{x}, t, \mathbf{y}) \hat{v}_1(\mathbf{x}, t) \det J(\cdot, \mathbf{y}) \, d\mathbf{x} \, dt \\ &+ \int_I \int_{D_{\text{ref}}} A(\mathbf{x}, \mathbf{y}) \nabla \hat{u}(\mathbf{x}, t, \mathbf{y}) \cdot \nabla \hat{v}_1(\mathbf{x}, t) \, d\mathbf{x} \, dt \\ &+ \int_{D_{\text{ref}}} \hat{u}(\mathbf{x}, 0, \mathbf{y}) \hat{v}_2(\mathbf{x}) \det J(\mathbf{x}, \mathbf{y}) \, d\mathbf{x} \end{aligned}$$

and

$$\begin{aligned} F(\mathbf{y}; \hat{v}) &:= \int_I \int_{D_{\text{ref}}} f_{\text{ref}}(\mathbf{x}, \mathbf{y}) \hat{v}_1(\mathbf{x}, t) \, d\mathbf{x} \, dt \\ &+ \int_{D_{\text{ref}}} \hat{u}_0(\mathbf{x}, \mathbf{y}) \hat{v}_2(\mathbf{x}) \det J(\mathbf{x}, \mathbf{y}) \, d\mathbf{x} \end{aligned}$$

and we denote

$$\begin{aligned} f_{\text{ref}}(\mathbf{x}, \mathbf{y}) &= f(\mathbf{V}(\mathbf{x}, \mathbf{y})) \det J(\mathbf{x}, \mathbf{y}), \\ \hat{u}_0(\mathbf{x}, \mathbf{y}) &= u_0(\mathbf{V}(\mathbf{x}, \mathbf{y}), \mathbf{y}), \\ A(\mathbf{x}, \mathbf{y}) &= (J(\mathbf{x}, \mathbf{y})^\top J(\mathbf{x}, \mathbf{y}))^{-1} \det J(\mathbf{x}, \mathbf{y}) \quad \text{and} \\ \mathcal{V} &:= L^2(I; H_0^1(D_{\text{ref}})) \times L^2(D_{\text{ref}}). \end{aligned}$$

We remark that $\hat{u}_0(\cdot, \mathbf{y})$ agrees with $\hat{u}(\cdot, 0, \mathbf{y})$ almost everywhere in D_{ref} .

Moreover, we define the space

$$\mathcal{X} := \{v \in L^2(I; H_0^1(D_{\text{ref}})) \mid \frac{\partial}{\partial t} v \in L^2(I; H^{-1}(D_{\text{ref}}))\} \quad (2.4)$$

endowed with the graph norm

$$\|v\|_{\mathcal{X}} := \left(\int_I (\|v(\cdot, t)\|_{H_0^1(D_{\text{ref}})}^2 + \|\frac{\partial}{\partial t} v(\cdot, t)\|_{H^{-1}(D_{\text{ref}})}^2) dt \right)^{1/2}.$$

Furthermore, we denote $\mathcal{L}^2 := L^2(D_{\text{ref}} \times I)$.

For our analysis of the pullback heat equation, we need to impose an additional conditions on the regularity of the initial condition u_0 as well as on the domain mapping. Hence, we state the following assumptions exclusively for this setting:

Assumptions.

(A8) *The initial condition $u_0: \mathcal{D} \rightarrow \mathbb{R}$ is infinitely many times continuously differentiable and satisfies*

$$\|\partial_{\mathbf{x}}^{\mathbf{v}} u_0\|_{L^2(\mathcal{D})} \leq C_{u_0} \quad \text{for all } \mathbf{v} \in \mathbb{N}_0^d$$

for some constant $C_{u_0} > 0$.

(A9) *There exist a constant $C \geq 1$, an exponent $\beta \geq 1$, and a sequence $\mathbf{b} = (b_j)_{j \geq 1}$ of non-negative numbers such that*

$$\|\partial_{\mathbf{y}}^{\mathbf{v}} \mathbf{V}(\cdot, \mathbf{y})\|_{W^{2,\infty}(D_{\text{ref}})} \leq C(|\mathbf{v}|!)^\beta \mathbf{b}^{\mathbf{v}} \quad \text{for all } \mathbf{v} \in \mathcal{F}, \mathbf{y} \in U.$$

From this condition it follows directly that

$$\|\partial_{\mathbf{y}}^{\mathbf{v}} J(\cdot, \mathbf{y})\|_{W^{1,\infty}(D_{\text{ref}})} \leq C(|\mathbf{v}|!)^\beta \mathbf{b}^{\mathbf{v}} \quad \text{for all } \mathbf{v} \in \mathcal{F}, \mathbf{y} \in U.$$

(A10) *There holds $\partial_t \hat{u}(\cdot, \cdot, \mathbf{y}) \in \mathcal{L}^2$ for all $\mathbf{y} \in U$.*

We note that the regularity analysis for the parabolic problem can also be carried out with the time derivative belonging to $L^2(I; H^{-1}(D_{\text{ref}}))$, since this still yields the regularity needed for the parametric derivatives of the solution (cf. Lemma 22). However, for clarity of presentation, we carry out the analysis under Assumption (A10).

We will show that under these assumptions, the pullback $\hat{u}_0 = u_0 \circ \mathbf{V}(\cdot, \mathbf{y})$ will belong to a Gevrey class and that for the parametric derivatives of the pullback solution there holds $\partial_{\mathbf{y}}^{\mathbf{v}} \hat{u} \in L^2(I; H_0^1(D_{\text{ref}}))$ for all $\mathbf{v} \in \mathcal{F}$, allowing us to complete the parametric regularity analysis for the parabolic model problem.

In what follows, we study QMC approximations of the expected value

$$\mathbb{E}[\hat{u}] = \int_U \hat{u}(\mathbf{y}) d\mathbf{y},$$

where $\hat{u}(\cdot, \cdot, \mathbf{y}) \in H_0^1(D_{\text{ref}})$ if $\hat{u}$ is the solution of the pullback Poisson equation (2.2) and $\hat{u}(\cdot, \cdot, \mathbf{y}) \in L^2(I; H_0^1(D_{\text{ref}}))$ if $\hat{u}$ is the solution of the pullback heat equation (2.3).

3. Quasi-Monte Carlo integration

Let $U_s := [-\frac{1}{2}, \frac{1}{2}]^s$. Consider the task of integrating a continuous function $F : U_s \rightarrow \mathbb{R}$ over an s -dimensional domain U_s

$$I_s(F) := \int_{U_s} F(\mathbf{y}) d\mathbf{y}.$$

A randomly shifted lattice rule (cf., e.g., [Dick et al. \(2013, 2022\)](#); [Kuo and Nuyens \(2016a, 2018\)](#)) is a method for QMC integration, represented as:

$$Q_{\text{ran}}(F) := \frac{1}{R} \sum_{r=1}^R Q_r(F), \quad \text{with } Q_r(F) := \frac{1}{n} \sum_{i=1}^n F(\{\mathbf{t}_i + \Delta^{(r)}\} - \frac{1}{2}), \quad (3.1)$$

where $\Delta^{(1)}, \dots, \Delta^{(R)}$ are i.i.d. random shifts drawn from $\mathcal{U}([0, 1]^s)$, $\{\cdot\}$ denotes the componentwise fractional part, and the lattice points are given by

$$\mathbf{t}_i := \left\{ \frac{i\mathbf{z}}{n} \right\}, \quad i \in \{1, \dots, n\},$$

where $\mathbf{z} \in \{0, \dots, n-1\}^s$ denotes the generating vector. Assume that the function F belongs to a weighted unanchored Sobolev space (cf., e.g., [Dick et al. \(2013, 2022\)](#); [Kuo and Nuyens \(2016a, 2018\)](#)) of dominating first-order smoothness $\mathcal{W}_{s, \boldsymbol{\gamma}} = \text{cl}_{\|\cdot\|_{\mathcal{W}_{s, \boldsymbol{\gamma}}}}(\mathcal{C}^\infty(U_s))$ equipped with the norm

$$\|F\|_{\mathcal{W}_{s, \boldsymbol{\gamma}}} := \left(\sum_{\mathbf{u} \subset \{1, \dots, s\}} \frac{1}{\gamma_{\mathbf{u}}} \int_{U_{|\mathbf{u}|}} \left(\int_{U_{s-|\mathbf{u}|}} \frac{\partial^{|\mathbf{u}|}}{\partial \mathbf{y}_{\mathbf{u}}} F(\mathbf{y}) d\mathbf{y}_{-\mathbf{u}} \right)^2 d\mathbf{y}_{\mathbf{u}} \right)^{1/2},$$

where $\boldsymbol{\gamma} := (\gamma_{\mathbf{u}})_{\mathbf{u} \subset \{1, \dots, s\}}$ is a collection of positive weights, $\mathbf{y}_{\mathbf{u}} := (y_j)_{j \in \mathbf{u}}$, and $\mathbf{y}_{-\mathbf{u}} := (y_j)_{j \in \{1, \dots, s\} \setminus \mathbf{u}}$. Then there exists a sequence of generating vectors using a component-by-component (CBC) algorithm ([Cools et al., 2006](#); [Dick et al., 2013](#); [Nuyens and Cools, 2006](#)) with the following error bound on the integration error.

Lemma 2 (cf. [Kuo and Nuyens \(2016a, Theorem 5.1\)](#)). *Let $F \in \mathcal{W}_{s, \boldsymbol{\gamma}}$ with positive weights $\boldsymbol{\gamma} = (\gamma_{\mathbf{u}})_{\mathbf{u} \subset \{1, \dots, s\}}$. An s -dimensional randomly shifted lattice rule with $n = 2^m$, $m \in \mathbb{N}$, points can be constructed by a CBC algorithm such that, for R independent random shifts and for all $\lambda \in (1/2, 1]$:*

$$\sqrt{\mathbb{E}_{\Delta}[|I_s(F) - Q_{\text{ran}}(F)|^2]} \leq \frac{1}{\sqrt{R}} \left(\frac{2}{n} \sum_{\emptyset \neq \mathbf{u} \subset \{1, \dots, s\}} \gamma_{\mathbf{u}}^\lambda \left(\frac{2\zeta(2\lambda)}{(2\pi^2)^\lambda} \right)^{|\mathbf{u}|} \right)^{\frac{1}{2\lambda}} \|F\|_{\mathcal{W}_{s, \boldsymbol{\gamma}}},$$

where $\mathbb{E}_{\Delta}[\cdot]$ denotes the expected value with respect to uniformly distributed random shifts over $[0, 1]^s$ and $\zeta(x) := \sum_{k=1}^{\infty} k^{-x}$ is the Riemann zeta function for $x > 1$.

4. Parametric regularity analysis

Our first task is to analyze $\partial_{\mathbf{y}}^{\mathbf{v}} A(\mathbf{x}, \mathbf{y})$ since it appears in the variational formulations for both the stationary and time-dependent model problems. We do this step-by-step:

- Step 1:* We first derive a regularity estimate for $\partial_{\mathbf{y}}^{\mathbf{v}}(J(\mathbf{x}, \mathbf{y})^T J(\mathbf{x}, \mathbf{y}))$.
Step 2: Step 1 will be used to deduce a bound for $\partial_{\mathbf{y}}^{\mathbf{v}}(J(\mathbf{x}, \mathbf{y})^T J(\mathbf{x}, \mathbf{y}))^{-1}$.
Step 3: We derive a regularity bound for $\partial_{\mathbf{y}}^{\mathbf{v}} \det J(\mathbf{x}, \mathbf{y})$.
Step 4: Finally, we use the Leibniz product rule to combine steps 2 and 3 to obtain a bound for $\partial_{\mathbf{y}}^{\mathbf{v}}((J(\mathbf{x}, \mathbf{y})^T J(\mathbf{x}, \mathbf{y}))^{-1} \det J(\mathbf{x}, \mathbf{y}))$.

This analysis is carried out in Subsection 4.1

We shall also be interested in the parametric regularity of the pullback source term f_{ref} . Our approach is as follows:

- Step 5:* We use Faà di Bruno's formula to obtain a regularity bound for $f(\mathbf{V}(\mathbf{x}, \mathbf{y}))$.
Step 6: The Leibniz product rule is used in conjunction with step 5 to obtain a bound for $\partial_{\mathbf{y}}^{\mathbf{v}} f_{\text{ref}}(\mathbf{x}, \mathbf{y})$.
 Parametric regularity analysis for the right-hand side will be carried out in Subsection 4.2.

4.1. Parametric regularity of the pullback coefficient

Step 1. We start by proving the following result.

Lemma 3. *Under Assumptions (A1) and (A3), there holds*

$$\|\partial_{\mathbf{y}}^{\mathbf{v}}(J(\cdot, \mathbf{y})^T J(\cdot, \mathbf{y}))\|_{L^\infty(D_{\text{ref}})} \leq C^2((|\mathbf{v}| + 1)!)^\beta \mathbf{b}^{\mathbf{v}}$$

for all $\mathbf{y} \in U$ and $\mathbf{v} \in \mathcal{F}$.

Proof. By the Leibniz product rule and Assumption (A3), we obtain for any $\mathbf{v} \in \mathcal{F}$ that

$$\begin{aligned} \|\partial_{\mathbf{y}}^{\mathbf{v}}(J(\cdot, \mathbf{y})^T J(\cdot, \mathbf{y}))\|_{L^\infty} &= \left\| \sum_{\mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} \partial_{\mathbf{y}}^{\mathbf{m}} J(\cdot, \mathbf{y})^T \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} J(\cdot, \mathbf{y}) \right\|_{L^\infty} \\ &\leq C^2 \mathbf{b}^{\mathbf{v}} \sum_{\mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} (|\mathbf{m}|!)^\beta (|\mathbf{v}-\mathbf{m}|!)^\beta \\ &= C^2 \mathbf{b}^{\mathbf{v}} \sum_{\ell=0}^{|\mathbf{v}|} (\ell!)^\beta ((|\mathbf{v}|-\ell)!)^\beta \sum_{\substack{\mathbf{m} \leq \mathbf{v} \\ |\mathbf{m}|=\ell}} \binom{\mathbf{v}}{\mathbf{m}} \\ &= C^2 \mathbf{b}^{\mathbf{v}} \sum_{\ell=0}^{|\mathbf{v}|} (\ell!)^\beta ((|\mathbf{v}|-\ell)!)^\beta \binom{|\mathbf{v}|}{\ell} \\ &= C^2 |\mathbf{v}|! \mathbf{b}^{\mathbf{v}} \sum_{\ell=0}^{|\mathbf{v}|} (\ell!)^{\beta-1} ((|\mathbf{v}|-\ell)!)^{\beta-1} \\ &\leq C^2 (|\mathbf{v}|!)^\beta \mathbf{b}^{\mathbf{v}} (|\mathbf{v}|+1) \\ &\leq C^2 ((|\mathbf{v}|+1)!)^\beta \mathbf{b}^{\mathbf{v}}, \end{aligned}$$

where we used the Vandermonde convolution $\sum_{\mathbf{m} \leq \mathbf{v}, |\mathbf{m}|=\ell} \binom{\mathbf{v}}{\mathbf{m}} = \binom{|\mathbf{v}|}{\ell}$ (cf., e.g., Quaintance and Gould (2015, Equation (5.1))) and the inequality $\ell!(v-\ell)! \leq v!$ which holds for all integers $0 \leq \ell \leq v$. $\square$

Step 2. Let $B(\mathbf{x}, \mathbf{y}) := J(\mathbf{x}, \mathbf{y})^T J(\mathbf{x}, \mathbf{y})$. Since

$$B(\mathbf{x}, \mathbf{y})B(\mathbf{x}, \mathbf{y})^{-1} = I_{d \times d},$$

we can use the Leibniz product rule to obtain, for any $\mathbf{v} \in \mathcal{F} \setminus \{\mathbf{0}\}$, that

$$\sum_{\mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} \partial_{\mathbf{y}}^{\mathbf{m}} B(\mathbf{x}, \mathbf{y}) \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} B(\mathbf{x}, \mathbf{y})^{-1} = 0.$$

Separating out the $\mathbf{m} = \mathbf{0}$ term, we obtain

$$\partial_{\mathbf{y}}^{\mathbf{v}} B(\mathbf{x}, \mathbf{y})^{-1} = -B(\mathbf{x}, \mathbf{y})^{-1} \sum_{\mathbf{0} \neq \mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} \partial_{\mathbf{y}}^{\mathbf{m}} B(\mathbf{x}, \mathbf{y}) \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} B(\mathbf{x}, \mathbf{y})^{-1}.$$

By Assumptions (A1)–(A3), there holds $\|B(\cdot, \mathbf{y})^{-1}\|_{L^\infty} \leq \sigma_{\min}^{-2}$ and we obtain

$$\|\partial_{\mathbf{y}}^{\mathbf{v}} B(\cdot, \mathbf{y})^{-1}\|_{L^\infty} \leq C^2 \sigma_{\min}^{-2} \sum_{\mathbf{0} \neq \mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} ((|\mathbf{m}| + 1)!)^\beta \mathbf{b}^{\mathbf{m}} \|\partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} B(\cdot, \mathbf{y})^{-1}\|_{L^\infty}.$$

Note that we have now expressed the $\mathbf{v}^{\text{th}}$ order partial derivative in terms of the lower order partial derivatives. Defining $\Lambda_{\mathbf{v}} := \|\partial_{\mathbf{y}}^{\mathbf{v}} B(\cdot, \mathbf{y})^{-1}\|_{L^\infty}$ yields the multidimensional recurrence relation

$$\Lambda_{\mathbf{0}} \leq \sigma_{\min}^{-2}, \tag{4.1}$$

$$\Lambda_{\mathbf{v}} \leq C^2 \sigma_{\min}^{-2} \sum_{\mathbf{0} \neq \mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} ((|\mathbf{m}| + 1)!)^\beta \mathbf{b}^{\mathbf{m}} \Lambda_{\mathbf{v}-\mathbf{m}} \quad \text{for all } \mathbf{v} \in \mathcal{F} \setminus \{\mathbf{0}\}. \tag{4.2}$$

This recursion admits the following explicit bound.

Lemma 4. *Let $(\Lambda_{\mathbf{v}})_{\mathbf{v} \in \mathcal{F}}$ be a sequence satisfying (4.1)–(4.2). Then there holds*

$$\Lambda_{\mathbf{v}} \leq \sigma_{\min}^{-2|\mathbf{v}|-2} C^{2|\mathbf{v}|} a_{\mathbf{v}} \mathbf{b}^{\mathbf{v}} \quad \text{for all } \mathbf{v} \in \mathcal{F},$$

where

$$a_{\mathbf{0}} = 1 \quad \text{and} \quad a_{\mathbf{v}} = \sum_{\substack{\mathbf{m} \leq \mathbf{v} \\ \mathbf{m} \neq \mathbf{0}}} \binom{\mathbf{v}}{\mathbf{m}} ((|\mathbf{m}| + 1)!)^\beta a_{\mathbf{v}-\mathbf{m}} \quad \text{for all } \mathbf{v} \in \mathcal{F} \setminus \{\mathbf{0}\}.$$

Proof. We prove the claim by induction with respect to the order of the multi-index $\mathbf{v}$. The basis of the induction $\mathbf{v} = \mathbf{0}$ follows immediately. We fix $\mathbf{v} \in \mathcal{F} \setminus \{\mathbf{0}\}$ and assume that the claim holds for all multi-indices with order less than $|\mathbf{v}|$. Then

$$\Lambda_{\mathbf{v}} \leq C^2 \sigma_{\min}^{-2} \sum_{\mathbf{0} \neq \mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} ((|\mathbf{m}| + 1)!)^\beta \mathbf{b}^{\mathbf{m}} \Lambda_{\mathbf{v}-\mathbf{m}}$$

$$\begin{aligned}
&\leq C^2 \sigma_{\min}^{-2} \sum_{\mathbf{0} \neq \mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} ((|\mathbf{m}| + 1)!)^\beta \mathbf{b}^{\mathbf{m}} \sigma_{\min}^{-2|\mathbf{v}|+2|\mathbf{m}|-2} \\
&\quad \times C^{2|\mathbf{v}|-2|\mathbf{m}|} a_{\mathbf{v}-\mathbf{m}} \mathbf{b}^{\mathbf{v}-\mathbf{m}} \\
&\leq C^{2|\mathbf{v}|} \sigma_{\min}^{-2|\mathbf{v}|-2} \mathbf{b}^{\mathbf{v}} \sum_{\mathbf{0} \neq \mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} ((|\mathbf{m}| + 1)!)^\beta a_{\mathbf{v}-\mathbf{m}} \\
&= C^{2|\mathbf{v}|} \sigma_{\min}^{-2|\mathbf{v}|-2} \mathbf{b}^{\mathbf{v}} a_{\mathbf{v}},
\end{aligned}$$

as desired. $\square$

By Lemmata 18–19, we conclude the following.

Lemma 5. *Under Assumptions (A1)–(A3), there holds*

$$\|\partial_{\mathbf{y}}^{\mathbf{v}} B(\cdot, \mathbf{y})^{-1}\|_{L^\infty(D_{\text{ref}})} \leq \sigma_{\min}^{-2|\mathbf{v}|-2} C^{2|\mathbf{v}|} (|\mathbf{v}|!)^\beta 4^{\beta|\mathbf{v}|} \max\{1, 2^{|\mathbf{v}|-1}\} \mathbf{b}^{\mathbf{v}}$$

for all $\mathbf{v} \in \mathcal{F}$ and $\mathbf{y} \in U$.

Step 3. We begin by proving an auxiliary result needed for the regularity bound of $\partial_{\mathbf{y}}^{\mathbf{v}} \det J(\mathbf{x}, \mathbf{y})$.

Lemma 6. *Under Assumptions (A1)–(A3), there holds*

$$\|\partial_{\mathbf{y}}^{\mathbf{v}} J(\cdot, \mathbf{y})^{-1}\|_{L^\infty(D_{\text{ref}})} \leq C^{|\mathbf{v}|} \sigma_{\min}^{-|\mathbf{v}|-1} \max\{1, 2^{|\mathbf{v}|-1}\} (|\mathbf{v}|!)^\beta \mathbf{b}^{\mathbf{v}}$$

for all $\mathbf{y} \in U$ and $\mathbf{v} \in \mathcal{F}$.

Proof. Let $\mathbf{v} \in \mathcal{F} \setminus \{\mathbf{0}\}$. Since

$$J(\mathbf{x}, \mathbf{y}) J(\mathbf{x}, \mathbf{y})^{-1} = I_{d \times d},$$

we obtain by the Leibniz product rule that

$$\sum_{\mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} \partial_{\mathbf{y}}^{\mathbf{m}} J(\mathbf{x}, \mathbf{y}) \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} J(\mathbf{x}, \mathbf{y})^{-1} = 0,$$

which is equivalent to

$$\partial_{\mathbf{y}}^{\mathbf{v}} J(\mathbf{x}, \mathbf{y})^{-1} = -J(\mathbf{x}, \mathbf{y})^{-1} \sum_{\mathbf{0} \neq \mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} \partial_{\mathbf{y}}^{\mathbf{m}} J(\mathbf{x}, \mathbf{y}) \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} J(\mathbf{x}, \mathbf{y})^{-1}.$$

It follows that

$$\begin{aligned}
&\|\partial_{\mathbf{y}}^{\mathbf{v}} J(\cdot, \mathbf{y})^{-1}\|_{L^\infty} \\
&\leq \|J(\cdot, \mathbf{y})^{-1}\|_{L^\infty} \sum_{\mathbf{0} \neq \mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} \|\partial_{\mathbf{y}}^{\mathbf{m}} J(\cdot, \mathbf{y})\|_{L^\infty} \|\partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} J(\cdot, \mathbf{y})^{-1}\|_{L^\infty}.
\end{aligned}$$

Now using that $\|J(\cdot, \mathbf{y})^{-1}\|_{L^\infty} \leq \sigma_{\min}^{-1}$ and the fact that by Assumption (A3) we have $\|\partial_{\mathbf{y}}^{\mathbf{m}} J(\cdot, \mathbf{y})\|_{L^\infty} \leq C(|\mathbf{m}|!)^\beta \mathbf{b}^{\mathbf{m}}$, we find

$$\|\partial_{\mathbf{y}}^{\mathbf{v}} J(\cdot, \mathbf{y})^{-1}\|_{L^\infty} \leq C \sigma_{\min}^{-1} \sum_{\mathbf{0} \neq \mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} (|\mathbf{m}|!)^\beta \mathbf{b}^{\mathbf{m}} \|\partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} J(\cdot, \mathbf{y})^{-1}\|_{L^\infty}.$$

We define $\xi_{\mathbf{v}} := \|\partial_{\mathbf{y}}^{\mathbf{v}} J(\cdot, \mathbf{y})^{-1}\|_{L^\infty}$. Solving for $\|\partial_{\mathbf{y}}^{\mathbf{v}} J(\cdot, \mathbf{y})^{-1}\|_{L^\infty}$ in the regularity bound is equivalent to solving

$$\xi_{\mathbf{0}} \leq \sigma_{\min}^{-1} \tag{4.3}$$

$$\xi_{\mathbf{v}} \leq C \sigma_{\min}^{-1} \sum_{\mathbf{0} \neq \mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} (|\mathbf{m}|!)^\beta \mathbf{b}^{\mathbf{m}} \xi_{\mathbf{v}-\mathbf{m}}. \tag{4.4}$$

for all $\mathbf{v} \in \mathcal{F} \setminus \{\mathbf{0}\}$. The claim follows by an application of Lemma 20 from the Appendix. $\square$

Lemma 7. *Under Assumptions (A1)–(A3), there holds*

$$\|\partial_{\mathbf{y}}^{\mathbf{v}} \det J(\cdot, \mathbf{y})\|_{L^\infty(D_{\text{ref}})} \leq \sigma_{\max}^d C_{\det J}^{|\mathbf{v}|} \frac{((|\mathbf{v}| + d^2 - 1)!)^\beta}{((d^2 - 1)!)^\beta} \mathbf{b}^{\mathbf{v}},$$

for all $\mathbf{v} \in \mathcal{F}$, $\mathbf{y} \in U$, where

$$C_{\det J} = \frac{2^\beta C}{\sigma_{\min}}.$$

Proof. If $\mathbf{v} = \mathbf{0}$, then the claim clearly holds since by Assumption (A2) and the fact that

$$\det J = \prod_{\sigma \in \sigma(J)} \sigma$$

we have

$$\|\det J(\cdot, \mathbf{y})\|_{L^\infty} \leq \sigma_{\max}^d.$$

Next, we let $\mathbf{v} \in \mathcal{F}$ and $k \geq 1$. We will proceed by induction and assume that the claim has already been proved for all multi-indices with order $\leq |\mathbf{v}|$. We can use Jacobi's formula (cf., e.g., Magnus and Neudecker (2019, Theorem 8.1)) and the Leibniz product rule in conjunction to obtain

$$\begin{aligned} \partial_{\mathbf{y}}^{\mathbf{v} + \mathbf{e}_k} \det J(\mathbf{x}, \mathbf{y}) &= \partial_{\mathbf{y}}^{\mathbf{v}} (\det J(\mathbf{x}, \mathbf{y}) \text{tr}(J(\mathbf{x}, \mathbf{y})^{-1} \partial_{\mathbf{y}}^{\mathbf{e}_k} J(\mathbf{x}, \mathbf{y}))) \\ &= \sum_{\mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} \partial_{\mathbf{y}}^{\mathbf{m}} \det J(\mathbf{x}, \mathbf{y}) \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \text{tr}(J(\mathbf{x}, \mathbf{y})^{-1} \partial_{\mathbf{y}}^{\mathbf{e}_k} J(\mathbf{x}, \mathbf{y})), \end{aligned} \tag{4.5}$$

where $\mathbf{e}_k \in \mathcal{F}$ denotes the multi-index whose k^{th} entry is 1 and the other entries are 0. By Lemma 6, there holds

$$\|\partial_{\mathbf{y}}^{\mathbf{v}} J(\cdot, \mathbf{y})^{-1}\|_{L^\infty} \leq C^{|\mathbf{v}|} \sigma_{\min}^{-|\mathbf{v}|-1} \max\{1, 2^{|\mathbf{v}|-1}\} (|\mathbf{v}|!)^\beta \mathbf{b}^{\mathbf{v}}.$$

Hence

$$\text{tr}(J(\mathbf{x}, \mathbf{y})^{-1} \partial_{\mathbf{y}}^{\mathbf{e}_k} J(\mathbf{x}, \mathbf{y})) = \sum_{k=1}^d \sum_{\ell=1}^d [J(\mathbf{x}, \mathbf{y})^{-1}]_{k,\ell} [\partial_{\mathbf{y}}^{\mathbf{e}_k} J(\mathbf{x}, \mathbf{y})]_{\ell,k}$$

and an application of the Leibniz product rule reveals that

$$\begin{aligned}
& \|\partial_{\mathbf{y}}^{\mathbf{m}} \text{tr}(J(\cdot, \mathbf{y})^{-1} \partial_{\mathbf{y}}^{\mathbf{e}_k} J(\cdot, \mathbf{y}))\|_{L^\infty} \\
&= \left\| \sum_{k=1}^d \sum_{\ell=1}^d \sum_{\mathbf{w} \leq \mathbf{m}} \binom{\mathbf{m}}{\mathbf{w}} [\partial_{\mathbf{y}}^{\mathbf{w}} J(\cdot, \mathbf{y})^{-1}]_{k,\ell} [\partial_{\mathbf{y}}^{\mathbf{m}-\mathbf{w}+\mathbf{e}_k} J(\cdot, \mathbf{y})]_{\ell,k} \right\|_{L^\infty} \\
&\leq C^{|\mathbf{m}|+1} \sigma_{\min}^{-|\mathbf{m}|-1} d^2 \mathbf{b}^{\mathbf{m}+\mathbf{e}_k} \\
&\quad \times \sum_{\mathbf{w} \leq \mathbf{m}} \binom{\mathbf{m}}{\mathbf{w}} \max\{1, 2^{|\mathbf{w}|-1}\} (|\mathbf{w}|!)^\beta ((|\mathbf{m}| - |\mathbf{w}| + 1)!)^\beta \\
&\leq C^{|\mathbf{m}|+1} \sigma_{\min}^{-|\mathbf{m}|-1} d^2 (|\mathbf{m}|!)^\beta \mathbf{b}^{\mathbf{m}+\mathbf{e}_k} \sum_{\ell=0}^{|\mathbf{m}|} \max\{1, 2^{\ell-1}\} (|\mathbf{m}| - \ell + 1)^\beta.
\end{aligned}$$

Here, we can estimate

$$\begin{aligned}
\sum_{\ell=0}^{|\mathbf{m}|} \max\{1, 2^{\ell-1}\} (|\mathbf{m}| - \ell + 1)^\beta &\leq \left(\sum_{\ell=0}^{|\mathbf{m}|} \max\{1, 2^{\ell-1}\}^{1/\beta} (|\mathbf{m}| - \ell + 1) \right)^\beta \\
&\leq \left(\sum_{\ell=0}^{|\mathbf{m}|} \max\{1, 2^{\ell-1}\} (|\mathbf{m}| - \ell + 1) \right)^\beta \\
&= (2^{|\mathbf{m}|+1} - 1)^\beta \leq 2^{\beta(|\mathbf{m}|+1)},
\end{aligned}$$

where we used the inequality $\sum_k a_k \leq (\sum_k a_k^\lambda)^{1/\lambda}$ which holds for all $a_k \geq 0$ and $\lambda \in (0, 1]$ (cf., e.g., [Hardy et al. \(1934, Theorem 19\)](#)). This yields

$$\begin{aligned}
& \|\partial_{\mathbf{y}}^{\mathbf{m}} \text{tr}(J(\cdot, \mathbf{y})^{-1} \partial_{\mathbf{y}}^{\mathbf{e}_k} J(\cdot, \mathbf{y}))\|_{L^\infty} \\
&\leq C^{|\mathbf{m}|+1} \sigma_{\min}^{-|\mathbf{m}|-1} d^2 2^{\beta(|\mathbf{m}|+1)} (|\mathbf{m}|!)^\beta \mathbf{b}^{\mathbf{m}+\mathbf{e}_k}.
\end{aligned} \tag{4.6}$$

Now the induction hypothesis together with the inequality (4.6) imply that (4.5) can be bounded by

$$\begin{aligned}
& \|\partial_{\mathbf{y}}^{\mathbf{v}+\mathbf{e}_k} \det J(\cdot, \mathbf{y})\|_{L^\infty} \\
&\leq \sum_{\mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} \sigma_{\max}^d \left(\frac{2^\beta C}{\sigma_{\min}} \right)^{|\mathbf{m}|} \frac{((|\mathbf{m}| + d^2 - 1)!)^\beta}{((d^2 - 1)!)^\beta} \mathbf{b}^{\mathbf{m}} \\
&\quad \times C^{|\mathbf{v}|-|\mathbf{m}|+1} \sigma_{\min}^{-|\mathbf{v}|-|\mathbf{m}|-1} d^2 2^{\beta(|\mathbf{v}|-|\mathbf{m}|+1)} ((|\mathbf{v}| - |\mathbf{m}|)!)^\beta \mathbf{b}^{\mathbf{v}-\mathbf{m}+\mathbf{e}_k} \\
&= \sigma_{\max}^d C_{\det J}^{|\mathbf{v}|+1} \mathbf{b}^{\mathbf{v}+\mathbf{e}_k} \frac{d^2}{((d^2 - 1)!)^\beta} \\
&\quad \times \sum_{\ell=0}^{|\mathbf{v}|} ((\ell + d^2 - 1)!)^\beta ((|\mathbf{v}| - \ell)!)^\beta \sum_{\substack{\mathbf{m} \leq \mathbf{v} \\ |\mathbf{m}|=\ell}} \binom{\mathbf{v}}{\mathbf{m}} \\
&\leq \sigma_{\max}^d C_{\det J}^{|\mathbf{v}|+1} \mathbf{b}^{\mathbf{v}+\mathbf{e}_k} \frac{d^2}{((d^2 - 1)!)^\beta} (|\mathbf{v}|!)^\beta \sum_{\ell=0}^{|\mathbf{v}|} \frac{((\ell + d^2 - 1)!)^\beta}{(\ell!)^\beta}.
\end{aligned} \tag{4.7}$$

Here, we can estimate

$$\sum_{\ell=0}^{|\mathbf{v}|} \frac{((\ell + d^2 - 1)!)^\beta}{(\ell!)^\beta} \leq \left(\sum_{\ell=0}^{|\mathbf{v}|} \frac{(\ell + d^2 - 1)!}{\ell!} \right)^\beta.$$

The above sum is *Gosper-summable* (cf., e.g., [Petkovšek et al. \(1996\)](#)): by letting $t_\ell := \frac{(\ell + d^2 - 1)!}{\ell!}$ and $z_\ell := \frac{\ell}{d^2} \frac{(\ell + d^2 - 1)!}{\ell!}$, there holds $t_\ell = z_{\ell+1} - z_\ell$ and hence

$$\sum_{\ell=0}^{|\mathbf{v}|} \frac{(\ell + d^2 - 1)!}{\ell!} = z_{|\mathbf{v}|+1} - z_0 = \frac{1}{d^2} \frac{(|\mathbf{v}| + d^2)!}{|\mathbf{v}|!}. \quad (4.8)$$

Combining the estimates (4.7)–(4.8) yields

$$\|\partial_{\mathbf{y}}^{\mathbf{v} + \mathbf{e}_k} \det J(\cdot, \mathbf{y})\|_{L^\infty} \leq \sigma_{\max}^d C_{\det J}^{|\mathbf{v}|+1} \mathbf{b}^{\mathbf{v} + \mathbf{e}_k} \frac{((|\mathbf{v}| + d^2)!)^\beta}{((d^2 - 1)!)^\beta}$$

as desired. $\square$

Step 4. By combining the previous steps, we obtain the parametric regularity bound for the pullback diffusion coefficient.

Lemma 8. *Let $\mathbf{y} \in U$ and $\mathbf{v} \in \mathcal{F}$. Under Assumptions (A1)–(A3), there holds*

$$\|\partial_{\mathbf{y}}^{\mathbf{v}} A(\cdot, \mathbf{y})\|_{L^\infty(D_{\text{ref}})} \leq C_{A,1} C_{A,2}^{|\mathbf{v}|} \max\{1, 2^{|\mathbf{v}|-1}\} \frac{((|\mathbf{v}| + d^2)!)^\beta}{(d^2!)^\beta} \mathbf{b}^{\mathbf{v}},$$

where

$$C_{A,1} = \sigma_{\max}^d \sigma_{\min}^{-2} \quad \text{and} \quad C_{A,2} = C_{\det J} \sigma_{\min}^{-2} C^2 4^\beta.$$

Proof. By the Leibniz product rule, we obtain that

$$\begin{aligned} & \|\partial_{\mathbf{y}}^{\mathbf{v}} (B(\cdot, \mathbf{y})^{-1} \det J(\cdot, \mathbf{y}))\|_{L^\infty} \\ & \leq \sum_{\mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} \|\partial_{\mathbf{y}}^{\mathbf{m}} \det J(\cdot, \mathbf{y})\|_{L^\infty} \|\partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} B(\cdot, \mathbf{y})^{-1}\|_{L^\infty} \\ & \leq \frac{\sigma_{\max}^d}{((d^2 - 1)!)^\beta} \sum_{\mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} C_{\det J}^{|\mathbf{m}|} ((|\mathbf{m}| + d^2 - 1)!)^\beta \mathbf{b}^{\mathbf{m}} \\ & \quad \times \sigma_{\min}^{-2|\mathbf{v}|+2|\mathbf{m}|-2} C^{2|\mathbf{v}|-2|\mathbf{m}|} ((|\mathbf{v}| - |\mathbf{m}|)!)^\beta 4^{\beta|\mathbf{v}|-\beta|\mathbf{m}|} \\ & \quad \times \max\{1, 2^{|\mathbf{v}|-|\mathbf{m}|-1}\} \mathbf{b}^{\mathbf{v}-\mathbf{m}} \\ & \leq \frac{\sigma_{\max}^d}{((d^2 - 1)!)^\beta} C_{\det J}^{|\mathbf{v}|} \sigma_{\min}^{-2|\mathbf{v}|-2} \max\{1, C^2\}^{|\mathbf{v}|} 4^{\beta|\mathbf{v}|} \max\{1, 2^{|\mathbf{v}|-1}\} \mathbf{b}^{\mathbf{v}} \end{aligned}$$

$$\begin{aligned}
& \times \underbrace{\sum_{\mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} ((|\mathbf{m}| + d^2 - 1)!)^\beta (|\mathbf{v}| - |\mathbf{m}|!)^\beta}_{\leq (|\mathbf{v}|!)^\beta \frac{(|\mathbf{v}|+1)^\beta \cdot ((|\mathbf{v}|+d^2)!)^\beta}{d^{2\beta} \cdot (|\mathbf{v}|+1)^\beta} \text{ as above}} \\
& = \frac{\sigma_{\max}^d}{(d^2!)^\beta} C_{\det J}^{|\mathbf{v}|} \sigma_{\min}^{-2|\mathbf{v}|-2} \max\{1, C^2\}^{|\mathbf{v}|} 4^{\beta|\mathbf{v}|} \\
& \quad \times \max\{1, 2^{|\mathbf{v}|-1}\} ((|\mathbf{v}| + d^2)!)^\beta \mathbf{b}^{\mathbf{v}},
\end{aligned}$$

as desired. $\square$

4.2. Regularity of the right-hand side

Step 5. We also need to control the parametric regularity of the pullback source term. We have the following result.

Lemma 9. *Under Assumptions (A1)–(A4), there holds*

$$\|\partial_{\mathbf{y}}^{\mathbf{v}} f(\mathbf{V}(\cdot, \mathbf{y}))\|_{L^\infty(D_{\text{ref}})} \leq 2^{\beta|\mathbf{v}|-\beta} \max\{1, \|\boldsymbol{\rho}\|_{\ell^1/\beta}^{|\mathbf{v}|}\} C_f C^{|\mathbf{v}|} \mathbf{b}^{\mathbf{v}} (|\mathbf{v}|!)^\beta$$

for all $\mathbf{v} \in \mathcal{F} \setminus \{\mathbf{0}\}$ and $\mathbf{y} \in U$.

Proof. By Faà di Bruno's formula (Savits, 2006), there holds

$$\partial_{\mathbf{y}}^{\mathbf{v}} f(\mathbf{V}(\mathbf{x}, \mathbf{y})) = \sum_{\substack{\boldsymbol{\lambda} \in \mathbb{N}_0^d \\ 1 \leq |\boldsymbol{\lambda}| \leq |\mathbf{v}|}} \partial_{\mathbf{x}}^{\boldsymbol{\lambda}} f(\mathbf{x}) \Big|_{\mathbf{x}=\mathbf{V}(\mathbf{x}, \mathbf{y})} \alpha_{\mathbf{v}, \boldsymbol{\lambda}}(\mathbf{x}, \mathbf{y}) \quad \text{for all } \mathbf{v} \in \mathcal{F} \setminus \{\mathbf{0}\},$$

where the sequence $(\alpha_{\mathbf{v}, \boldsymbol{\lambda}}(\mathbf{x}, \mathbf{y}))_{\mathbf{v} \in \mathcal{F}, \boldsymbol{\lambda} \in \mathbb{Z}^d}$ is defined recursively by setting $\alpha_{\mathbf{v}, \mathbf{0}} \equiv \delta_{\mathbf{v}, \mathbf{0}}$ for all $\mathbf{v} \in \mathcal{F}$, $\alpha_{\mathbf{v}, \boldsymbol{\lambda}} \equiv 0$ if $|\mathbf{v}| < |\boldsymbol{\lambda}|$ or $\boldsymbol{\lambda} \not\geq \mathbf{0}$, and

$$\alpha_{\mathbf{v}+\mathbf{e}_j, \boldsymbol{\lambda}}(\mathbf{x}, \mathbf{y}) = \sum_{\ell \in \text{supp}(\boldsymbol{\lambda})} \sum_{0 \leq |\mathbf{m}| \leq |\mathbf{v}|} \binom{\mathbf{v}}{\mathbf{m}} \partial_{\mathbf{y}}^{\mathbf{m}+\mathbf{e}_j} [\mathbf{V}(\mathbf{x}, \mathbf{y})]_\ell \alpha_{\mathbf{v}-\mathbf{m}, \boldsymbol{\lambda}-\mathbf{e}_\ell}(\mathbf{x}, \mathbf{y})$$

otherwise, where $\mathbf{e}_j \in \mathcal{F}$ denotes the multi-index whose j^{th} component is 1 and other entries are 0.

By Assumption (A3), we especially have that

$$\begin{aligned}
& \|\alpha_{\mathbf{v}+\mathbf{e}_j, \boldsymbol{\lambda}}(\cdot, \mathbf{y})\|_{L^\infty} \\
& \leq C \sum_{\ell \in \text{supp}(\boldsymbol{\lambda})} \sum_{0 \leq |\mathbf{m}| \leq |\mathbf{v}|} \binom{\mathbf{v}}{\mathbf{m}} ((|\mathbf{m}| + 1)!)^\beta \mathbf{b}^{\mathbf{m}+\mathbf{e}_j} \|\alpha_{\mathbf{v}-\mathbf{m}, \boldsymbol{\lambda}-\mathbf{e}_\ell}(\cdot, \mathbf{y})\|_{L^\infty}.
\end{aligned}$$

As a consequence of Kaarnioja and Schillings (2024, Theorem 5.1), the sequence $\|\alpha_{\mathbf{v}, \boldsymbol{\lambda}}(\cdot, \mathbf{y})\|_{L^\infty}$ can be bounded by

$$\|\alpha_{\mathbf{v}, \boldsymbol{\lambda}}(\cdot, \mathbf{y})\|_{L^\infty} \leq C^{|\mathbf{v}|} \frac{(|\mathbf{v}|!)^\beta ((|\mathbf{v}| - 1)!)^\beta}{(\boldsymbol{\lambda}!)^\beta ((|\mathbf{v}| - |\boldsymbol{\lambda}|)!)^\beta ((|\boldsymbol{\lambda}| - 1)!)^\beta} \mathbf{b}^{\mathbf{v}}$$

and, by employing Assumption (A4), we obtain

$$\|\partial_{\mathbf{y}}^{\mathbf{v}} f(\mathbf{V}(\cdot, \mathbf{y}))\|_{L^\infty}$$

$$\begin{aligned}
 &\leq C_f C^{|\mathbf{v}|} \mathbf{b}^{\mathbf{v}} (|\mathbf{v}|!)^\beta ((|\mathbf{v}| - 1)!)^\beta \sum_{\substack{\boldsymbol{\lambda} \in \mathbb{N}_0^d \\ 1 \leq |\boldsymbol{\lambda}| \leq |\mathbf{v}|}} \frac{(|\boldsymbol{\lambda}|!)^\beta}{(|\boldsymbol{\lambda}|!)^\beta ((|\mathbf{v}| - |\boldsymbol{\lambda}|)!)^\beta ((|\boldsymbol{\lambda}| - 1)!)^\beta} \boldsymbol{\rho}^{\boldsymbol{\lambda}} \\
 &= C_f C^{|\mathbf{v}|} \mathbf{b}^{\mathbf{v}} (|\mathbf{v}|!)^\beta ((|\mathbf{v}| - 1)!)^\beta \sum_{\ell=1}^{|\mathbf{v}|} \frac{(\ell!)^\beta}{((|\mathbf{v}| - \ell)!)^\beta ((\ell - 1)!)^\beta} \sum_{\substack{\boldsymbol{\lambda} \in \mathbb{N}_0^d \\ |\boldsymbol{\lambda}| = \ell}} \frac{1}{(|\boldsymbol{\lambda}|!)^\beta} \boldsymbol{\rho}^{\boldsymbol{\lambda}} \\
 &\leq C_f C^{|\mathbf{v}|} \mathbf{b}^{\mathbf{v}} (|\mathbf{v}|!)^\beta ((|\mathbf{v}| - 1)!)^\beta \sum_{\ell=1}^{|\mathbf{v}|} \frac{1}{((|\mathbf{v}| - \ell)!)^\beta ((\ell - 1)!)^\beta} \left(\sum_{\substack{\boldsymbol{\lambda} \in \mathbb{N}_0^d \\ |\boldsymbol{\lambda}| = \ell}} \frac{\ell!}{(|\boldsymbol{\lambda}|!)^\beta} \boldsymbol{\rho}^{\boldsymbol{\lambda}/\beta} \right)^\beta \\
 &= C_f C^{|\mathbf{v}|} \mathbf{b}^{\mathbf{v}} (|\mathbf{v}|!)^\beta ((|\mathbf{v}| - 1)!)^\beta \sum_{\ell=1}^{|\mathbf{v}|} \frac{1}{((|\mathbf{v}| - \ell)!)^\beta ((\ell - 1)!)^\beta} \left(\sum_{j=1}^d \rho_j^{1/\beta} \right)^{\beta \ell} \\
 &\leq \max\{1, \|\boldsymbol{\rho}\|_{\ell^{1/\beta}}^{|\mathbf{v}|}\} C_f C^{|\mathbf{v}|} \mathbf{b}^{\mathbf{v}} (|\mathbf{v}|!)^\beta ((|\mathbf{v}| - 1)!)^\beta \sum_{\ell=1}^{|\mathbf{v}|} \frac{1}{((|\mathbf{v}| - \ell)!)^\beta ((\ell - 1)!)^\beta} \\
 &= 2^{\beta|\mathbf{v}| - \beta} \max\{1, \|\boldsymbol{\rho}\|_{\ell^{1/\beta}}^{|\mathbf{v}|}\} C_f C^{|\mathbf{v}|} \mathbf{b}^{\mathbf{v}} (|\mathbf{v}|!)^\beta,
 \end{aligned}$$

where we used the multinomial theorem as well as the following bound (cf., e.g., [Kaarnioja and Schillings \(2024, Lemma A.1\)](#))

$$\sum_{\ell=1}^v \frac{1}{((v - \ell)!)^\beta ((\ell - 1)!)^\beta} \leq \left(\sum_{\ell=1}^v \frac{1}{(v - \ell)! (\ell - 1)!} \right)^\beta = \frac{2^{\beta \ell - \beta}}{((v - 1)!)^\beta}.$$

This proves the assertion. $\square$

Lemma 10. *Let $\mathbf{v} \in \mathcal{F} \setminus \{\mathbf{0}\}$ and $\mathbf{y} \in U$. Under Assumptions (A1)–(A4), there holds*

$$\|\partial_{\mathbf{y}}^{\mathbf{v}} f_{\text{ref}}(\cdot, \mathbf{y})\|_{L^\infty(D_{\text{ref}})} \leq C_{f_{\text{ref}},1} C_{f_{\text{ref}},2}^{|\mathbf{v}|} \max\{1, \|\boldsymbol{\rho}\|_{\ell^{1/\beta}}^{|\mathbf{v}|}\} \frac{((|\mathbf{v}| + d^2)!)^\beta}{(d^2!)^\beta} \mathbf{b}^{\mathbf{v}},$$

where

$$C_{f_{\text{ref}},1} = \sigma_{\max}^d 2^{-\beta} C_f \quad \text{and} \quad C_{f_{\text{ref}},2} = \frac{2^\beta C^2}{\sigma_{\min}} 2^\beta.$$

Proof. The claim follows by an application of the Leibniz product rule:

$$\|\partial_{\mathbf{y}}^{\mathbf{v}} f_{\text{ref}}(\cdot, \mathbf{y})\|_{L^\infty} \leq \sum_{\mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} \|\partial_{\mathbf{y}}^{\mathbf{m}} \det J(\cdot, \mathbf{y})\|_{L^\infty} \|\partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} f(\mathbf{V}(\cdot, \mathbf{y}))\|_{L^\infty},$$

where the result follows using the same steps as the proof of Lemma 8. $\square$

4.3. Parametric regularity of the solution to the pullback Poisson equation

We have the following.

Theorem 11. *Let $\mathbf{v} \in \mathcal{F}$ and $\mathbf{y} \in U$. Under Assumptions (A1)–(A4), the solution to (2.2) satisfies*

$$\|\partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\cdot, \mathbf{y})\|_{H_0^1(D_{\text{ref}})} \leq C_{\widehat{u},1} C_{\widehat{u},2}^{|\mathbf{v}|} (|\mathbf{v}|!)^\beta \mathbf{b}^{\mathbf{v}}$$

where

$$C_{\widehat{u},1} = 1 + \frac{\sigma_{\max}^2}{\sigma_{\min}^d} |D_{\text{ref}}|^{1/2} C_{D_{\text{ref}}} C_{f_{\text{ref}},1},$$

$$C_{\widehat{u},2} = \max\{c_0, c_1, \tilde{c}_1, \tilde{c}_2\}^2 ((d^2)!)^\beta 2^{\beta(d^2+1)+1},$$

with

$$c_0 = \frac{C_{A,1}}{\sigma_{\min}^d ((d^2)!)^\beta}, \quad c_1 = 2C_{A,2}, \quad \tilde{c}_0 = \frac{|D_{\text{ref}}|^{1/2} C_{D_{\text{ref}}} C_{f_{\text{ref}},1}}{\sigma_{\min}^d ((d^2)!)^\beta}, \quad \text{and}$$

$$\tilde{c}_1 = C_{f_{\text{ref}},2} \max\{1, \|\boldsymbol{\rho}\|_{\ell^{1/\beta}}\}.$$

Proof. By differentiating the weak formulation (2.2) on both sides with respect to $\mathbf{y} \in U$, we obtain

$$\sum_{\mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{m}} A(\mathbf{x}, \mathbf{y}) \nabla \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \widehat{u}(\mathbf{x}, \mathbf{y}) \cdot \nabla \widehat{v}(\mathbf{x}) \, d\mathbf{x}$$

$$= \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{v}} f_{\text{ref}}(\cdot, \mathbf{y}) \widehat{v}(\mathbf{x}) \, d\mathbf{x}.$$

By using standard estimates and testing again with $\widehat{v} = \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\cdot, \mathbf{y}) \in H_0^1(D_{\text{ref}})$, we obtain the recurrence relation

$$\|\partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\cdot, \mathbf{y})\|_{H_0^1} \leq \frac{1}{\sigma_{\min}^d} \sum_{\substack{\mathbf{m} \leq \mathbf{v} \\ \mathbf{m} \neq \mathbf{0}}} \binom{\mathbf{v}}{\mathbf{m}} \|\partial_{\mathbf{y}}^{\mathbf{m}} A(\cdot, \mathbf{y})\|_{L^\infty} \|\partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \widehat{u}(\cdot, \mathbf{y})\|_{H_0^1}$$

$$+ \frac{|D_{\text{ref}}|^{1/2} C_{D_{\text{ref}}}}{\sigma_{\min}^d} \|\partial_{\mathbf{y}}^{\mathbf{v}} f_{\text{ref}}\|_{L^\infty},$$

where $|D_{\text{ref}}| = \int_{D_{\text{ref}}} d\mathbf{x}$ and $C_{D_{\text{ref}}} > 0$ denotes the Poincaré constant of domain D_{ref} .

We write the parametric regularity bound for $A(\mathbf{x}, \mathbf{y})$ as

$$\|\partial_{\mathbf{y}}^{\mathbf{v}} A(\cdot, \mathbf{y})\|_{L^\infty} \leq \sigma_{\min}^d c_0 c_1^{|\mathbf{v}|} ((|\mathbf{v}| + d^2)!)^\beta \mathbf{b}^{\mathbf{v}},$$

where $c_0 = \frac{C_{A,1}}{\sigma_{\min}^d ((d^2)!)^\beta}$ and $c_1 = 2C_{A,2}$, and the parametric regularity bound for $f_{\text{ref}}(\mathbf{x}, \mathbf{y}) = f(\mathbf{V}(\mathbf{x}, \mathbf{y})) \det J(\mathbf{x}, \mathbf{y})$ as

$$\|\partial_{\mathbf{y}}^{\mathbf{v}} f_{\text{ref}}(\cdot, \mathbf{y})\|_{L^\infty} \leq \frac{\sigma_{\min}^d \tilde{c}_0}{|D_{\text{ref}}|^{1/2} C_{D_{\text{ref}}}} \tilde{c}_1^{|\mathbf{v}|} ((|\mathbf{v}| + d^2)!)^\beta \mathbf{b}^{\mathbf{v}},$$

where $\tilde{c}_0 = \frac{|D_{\text{ref}}|^{1/2} C_{D_{\text{ref}}} C_{f_{\text{ref}},1}}{\sigma_{\min}^d ((d^2)!)^\beta}$ and $\tilde{c}_1 = C_{f_{\text{ref}},2} \max\{1, \|\boldsymbol{\rho}\|_{\ell^{1/\beta}}\}$. Then we obtain the recursive bound

$$\begin{aligned} \|\hat{u}(\cdot, \mathbf{y})\|_{H_0^1} &\leq C_0, \\ \|\partial_{\mathbf{y}}^{\mathbf{v}} \hat{u}(\cdot, \mathbf{y})\|_{H_0^1} &\leq C \sum_{\substack{\mathbf{m} \leq \mathbf{v} \\ \mathbf{m} \neq \mathbf{0}}} \binom{\mathbf{v}}{\mathbf{m}} C^{|\mathbf{m}|} ((|\mathbf{m}| + d^2)!)^\beta \mathbf{b}^{\mathbf{m}} \|\partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \hat{u}(\cdot, \mathbf{y})\|_{H_0^1} \\ &\quad + C C^{|\mathbf{v}|} ((|\mathbf{v}| + d^2)!)^\beta \mathbf{b}^{\mathbf{v}}, \end{aligned}$$

where $C_0 = \frac{\sigma_{\max}^2}{\sigma_{\min}^d} |D_{\text{ref}}|^{1/2} C_{D_{\text{ref}}} C_{f_{\text{ref}},1}$ and $C = \max\{c_0, c_1, \tilde{c}_0, \tilde{c}_1\}$. By Lemma 21, we obtain the inductive bound

$$\|\partial_{\mathbf{y}}^{\mathbf{v}} \hat{u}(\cdot, \mathbf{y})\|_{H_0^1} \leq (1 + C_0) C^{|\mathbf{v}|} (|\mathbf{v}|!)^\beta \tilde{\tau}_{|\mathbf{v}|, \beta, d^2} \mathbf{b}^{\mathbf{v}},$$

where $\tilde{\tau}$ is defined by the recursion (7.4). Especially, we can bound

$$\tilde{\tau}_{\mathbf{v}, \beta, d^2} \leq C^{\mathbf{v}} \tau_{\mathbf{v}, \beta, d^2}$$

with τ defined by (7.1). The claim follows by an application of Lemma 19. $\square$

4.4. Parametric regularity of the solution to the pullback heat equation

In contrast to the pullback Poisson equation, the pullback heat equation contains several additional terms which need to be accounted for further complicating the analysis. Thus we approach the analysis step-by-step as follows:

- Step 1:* We begin by differentiating the variational formulation (2.3) with respect to the parametric variable.
- Step 2:* In order to obtain bounds for the individual terms, we need to fix the test function satisfying certain conditions.
- Step 3:* Using steps 1 and 2, we derive a lower bound on the left-hand side of the differentiated variational formulation.
- Step 4:* We proceed to derive an upper bound for the right-hand side of the differentiated variational formulation.
- Step 5:* We shall additionally require a parametric regularity bound for the pullback initial condition $\hat{u}_0$.
- Step 6:* Finally, we combine the bounds derived in steps 3–5 and derive an inductive bound for the solution to the pullback heat equation.

Step 1. As shown in Lemma 22 we have that $\partial_{\mathbf{y}}^{\mathbf{v}} \hat{u}(\cdot, \cdot, \mathbf{y}) \in L^2(I; H_0^1(D_{\text{ref}}))$ for all $\mathbf{v} \in \mathcal{F}$. By differentiating the variational formulation (2.3) on both sides with respect to $\mathbf{y}$ and applying the Leibniz product rule, we obtain

$$\begin{aligned} \sum_{\mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} &\left[\int_I \int_{D_{\text{ref}}} \frac{\partial}{\partial t} (\partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \hat{u}(\cdot, t, \mathbf{y})) \hat{v}_1(\cdot, t) \partial_{\mathbf{y}}^{\mathbf{m}} \det J(\cdot, \mathbf{y}) \, d\mathbf{x} \, dt \right. \\ &\quad \left. + \int_I \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{m}} A(\mathbf{x}, \mathbf{y}) \nabla \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \hat{u}(\mathbf{x}, t, \mathbf{y}) \cdot \nabla \hat{v}_1(\mathbf{x}, t) \, d\mathbf{x} \, dt \right] \end{aligned}$$

$$\begin{aligned}
& + \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \widehat{u}(\mathbf{x}, 0, \mathbf{y}) \widehat{v}_2(\mathbf{x}) \partial_{\mathbf{y}}^{\mathbf{m}} \det J(\mathbf{x}, \mathbf{y}) \, d\mathbf{x} \Big] \\
& = \int_I \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{v}} f_{\text{ref}}(\cdot, t, \mathbf{y}) \widehat{v}_1 \, d\mathbf{x} \, dt \\
& \quad + \sum_{\mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \widehat{u}_0(\mathbf{x}, \mathbf{y}) \widehat{v}_2(\mathbf{x}) \partial_{\mathbf{y}}^{\mathbf{m}} \det J(\mathbf{x}, \mathbf{y}) \, d\mathbf{x},
\end{aligned}$$

where $\widehat{u}_0(\mathbf{x}, \mathbf{y}) = u_0(\mathbf{V}(\mathbf{x}, \mathbf{y}), \mathbf{y})$ for $\mathbf{x} \in D_{\text{ref}}$, $\mathbf{y} \in U$. Separating out the $\mathbf{m} = \mathbf{0}$ term, we obtain

$$\begin{aligned}
& \int_I \int_{D_{\text{ref}}} \frac{\partial}{\partial t} (\partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\cdot, t, \mathbf{y})) \widehat{v}_1(\cdot, t) \det J(\cdot, \mathbf{y}) \, d\mathbf{x} \, dt \\
& \quad + \int_I \int_{D_{\text{ref}}} A(\mathbf{x}, \mathbf{y}) \nabla \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\mathbf{x}, t, \mathbf{y}) \cdot \nabla \widehat{v}_1(\mathbf{x}, t) \, d\mathbf{x} \, dt \\
& \quad + \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\mathbf{x}, 0, \mathbf{y}) \widehat{v}_2(\mathbf{x}) \det J(\mathbf{x}, \mathbf{y}) \, d\mathbf{x} \\
& = - \sum_{\mathbf{0} \neq \mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} \Big[\int_I \int_{D_{\text{ref}}} \frac{\partial}{\partial t} \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \widehat{u}(\cdot, t, \mathbf{y}) \widehat{v}_1(\cdot, t) \partial_{\mathbf{y}}^{\mathbf{m}} \det J(\cdot, \mathbf{y}) \, d\mathbf{x} \, dt \\
& \quad + \int_I \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{m}} A(\mathbf{x}, \mathbf{y}) \nabla \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \widehat{u}(\mathbf{x}, t, \mathbf{y}) \cdot \nabla \widehat{v}_1(\mathbf{x}, t) \, d\mathbf{x} \, dt \\
& \quad + \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \widehat{u}(\mathbf{x}, 0, \mathbf{y}) \widehat{v}_2(\mathbf{x}) \partial_{\mathbf{y}}^{\mathbf{m}} \det J(\mathbf{x}, \mathbf{y}) \, d\mathbf{x} \Big] \\
& \quad + \int_I \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{v}} f_{\text{ref}}(\cdot, t, \mathbf{y}) \widehat{v}_1 \, d\mathbf{x} \, dt \\
& \quad + \sum_{\mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \widehat{u}_0(\mathbf{x}, \mathbf{y}) \widehat{v}_2(\mathbf{x}) \partial_{\mathbf{y}}^{\mathbf{m}} \det J(\mathbf{x}, \mathbf{y}) \, d\mathbf{x}.
\end{aligned}$$

Step 2. Let $\mathcal{S}(\mathbf{y}) : H_0^1(D_{\text{ref}}) \rightarrow H^{-1}(D_{\text{ref}})$ denote the differential operator

$$\langle \mathcal{S}(\mathbf{y}) \widehat{v}, \widehat{w} \rangle_{H^{-1}(D_{\text{ref}}), H_0^1(D_{\text{ref}})} := \int_{D_{\text{ref}}} A(\mathbf{x}, \mathbf{y}) \nabla \widehat{v}(\mathbf{x}) \cdot \nabla \widehat{w}(\mathbf{x}) \, d\mathbf{x}.$$

We consider the test function

$$(\widehat{v}_1, \widehat{v}_2) = (\widehat{z}(\cdot, \cdot, \mathbf{y}) + \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\cdot, \cdot, \mathbf{y}), \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\mathbf{x}, 0, \mathbf{y})) \in \mathcal{V},$$

where $\widehat{z}(\mathbf{x}, t, \mathbf{y}) = (-\Delta)^{-1} (\det J(\mathbf{x}, \mathbf{y}) \frac{\partial}{\partial t} \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u})$ and the negative Laplacian

$$-\Delta : H_0^1(D_{\text{ref}}) \cap H^2(D_{\text{ref}}) \rightarrow L^2(D_{\text{ref}}).$$

Especially, there hold

$$\|\nabla v\|_{H_0^1(D_{\text{ref}})} \geq \frac{1}{C_{D_{\text{ref}}}} \|v\|_{L^2(D_{\text{ref}})}$$

and

$$\| -\Delta v \|_{L^2(D_{\text{ref}})} \leq \| v \|_{H_0^1(D_{\text{ref}}) \cap H^2(D_{\text{ref}})}$$

for all $v \in H_0^1(D_{\text{ref}}) \cap H^2(D_{\text{ref}})$.

Hence

$$\begin{aligned} \int_{D_{\text{ref}}} \frac{\partial}{\partial t} \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u} \widehat{z} \det J \, d\mathbf{x} &= \langle -\Delta \widehat{z}, \widehat{z} \rangle_{L^2(D_{\text{ref}})} = \langle \nabla \widehat{z}, \nabla \widehat{z} \rangle_{L^2(D_{\text{ref}})} \\ &\geq \frac{1}{C_{D_{\text{ref}}}^2} \|\widehat{z}\|_{H_0^1(D_{\text{ref}})}^2 \geq \frac{1}{C_{D_{\text{ref}}}^2} \|\det J(\cdot, \mathbf{y}) \frac{\partial}{\partial t} \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}\|_{L^2(D_{\text{ref}})}^2 \\ &\geq \frac{\sigma_{\min}^{2d}}{C_{D_{\text{ref}}}^2} \|\frac{\partial}{\partial t} \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}\|_{L^2(D_{\text{ref}})}^2; \end{aligned}$$

see also [Schwab and Stevenson \(2009, pg. 1315\)](#).

Plugging in the aforementioned test function, the variational formulation becomes

$$\begin{aligned} &\int_I \int_{D_{\text{ref}}} \frac{\partial}{\partial t} \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\mathbf{x}, t, \mathbf{y}) \widehat{z}(\mathbf{x}, t, \mathbf{y}) \det J(\cdot, \mathbf{y}) \, d\mathbf{x} \, dt \\ &\quad + \int_I \int_{D_{\text{ref}}} \frac{\partial}{\partial t} \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\mathbf{x}, t, \mathbf{y}) \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\mathbf{x}, t, \mathbf{y}) \det J(\cdot, \mathbf{y}) \, d\mathbf{x} \, dt \\ &\quad + \int_I \int_{D_{\text{ref}}} A(\mathbf{x}, \mathbf{y}) \nabla \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\mathbf{x}, t, \mathbf{y}) \cdot \nabla \widehat{z}(\mathbf{x}, t, \mathbf{y}) \, d\mathbf{x} \, dt \\ &\quad + \int_I \int_{D_{\text{ref}}} A(\mathbf{x}, \mathbf{y}) \nabla \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\mathbf{x}, t, \mathbf{y}) \cdot \nabla \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\mathbf{x}, t, \mathbf{y}) \, d\mathbf{x} \, dt \\ &\quad + \int_{D_{\text{ref}}} |\partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\mathbf{x}, 0, \mathbf{y})|^2 \det J(\mathbf{x}, \mathbf{y}) \, d\mathbf{x} \\ &= - \sum_{\mathbf{0} \neq \mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} \left[\int_I \int_{D_{\text{ref}}} \frac{\partial}{\partial t} \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \widehat{u}(\cdot, t, \mathbf{y}) \widehat{z}(\cdot, t, \mathbf{y}) \partial_{\mathbf{y}}^{\mathbf{m}} \det J(\cdot, \mathbf{y}) \, d\mathbf{x} \, dt \right. \\ &\quad + \int_I \int_{D_{\text{ref}}} \frac{\partial}{\partial t} \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \widehat{u}(\cdot, t, \mathbf{y}) \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\cdot, t, \mathbf{y}) \partial_{\mathbf{y}}^{\mathbf{m}} \det J(\cdot, \mathbf{y}) \, d\mathbf{x} \, dt \\ &\quad + \int_I \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{m}} A(\mathbf{x}, \mathbf{y}) \nabla \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \widehat{u}(\mathbf{x}, t, \mathbf{y}) \cdot \nabla \widehat{z}(\mathbf{x}, t, \mathbf{y}) \, d\mathbf{x} \, dt \\ &\quad + \int_I \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{m}} A(\mathbf{x}, \mathbf{y}) \nabla \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \widehat{u}(\mathbf{x}, t, \mathbf{y}) \cdot \nabla \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\mathbf{x}, t, \mathbf{y}) \, d\mathbf{x} \, dt \\ &\quad \left. + \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \widehat{u}(\mathbf{x}, 0, \mathbf{y}) \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\mathbf{x}, 0, \mathbf{y}) \partial_{\mathbf{y}}^{\mathbf{m}} \det J(\mathbf{x}, \mathbf{y}) \, d\mathbf{x} \right] \\ &\quad + \int_I \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{v}} f_{\text{ref}}(\cdot, t, \mathbf{y}) \widehat{z} \, d\mathbf{x} \, dt + \int_I \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{v}} f_{\text{ref}}(\cdot, t, \mathbf{y}) \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u} \, d\mathbf{x} \, dt \\ &\quad + \sum_{\mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \widehat{u}_0(\mathbf{x}, \mathbf{y}) \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\mathbf{x}, 0, \mathbf{y}) \partial_{\mathbf{y}}^{\mathbf{m}} \det J(\mathbf{x}, \mathbf{y}) \, d\mathbf{x}. \end{aligned}$$

Step 3. The left-hand side can be bounded from below by noting that, by definition of $\widehat{z}$, there holds

$$\int_{D_{\text{ref}}} A \nabla \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u} \cdot \nabla \widehat{z} \, d\mathbf{x} = \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u} \frac{\partial}{\partial t} \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u} \det J(\mathbf{x}, \mathbf{y}) \, d\mathbf{x},$$

and thus

$$\begin{aligned} & \int_I \int_{D_{\text{ref}}} \frac{\partial}{\partial t} \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\mathbf{x}, t, \mathbf{y}) \widehat{z}(\mathbf{x}, t, \mathbf{y}) \det J(\cdot, \mathbf{y}) \, d\mathbf{x} \, dt \\ & + \int_I \int_{D_{\text{ref}}} \frac{\partial}{\partial t} \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\mathbf{x}, t, \mathbf{y}) \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\mathbf{x}, t, \mathbf{y}) \det J(\cdot, \mathbf{y}) \, d\mathbf{x} \, dt \\ & + \int_I \int_{D_{\text{ref}}} A(\mathbf{x}, \mathbf{y}) \nabla \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\mathbf{x}, t, \mathbf{y}) \cdot \nabla \widehat{z}(\mathbf{x}, t, \mathbf{y}) \, d\mathbf{x} \, dt \\ & + \int_I \int_{D_{\text{ref}}} A(\mathbf{x}, \mathbf{y}) \nabla \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\mathbf{x}, t, \mathbf{y}) \cdot \nabla \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\mathbf{x}, t, \mathbf{y}) \, d\mathbf{x} \, dt \\ & + \int_{D_{\text{ref}}} |\partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\mathbf{x}, 0, \mathbf{y})|^2 \det J(\mathbf{x}, \mathbf{y}) \, d\mathbf{x} \\ & \geq \frac{\sigma_{\min}^{2d}}{C_{D_{\text{ref}}}^2} \left\| \frac{\partial}{\partial t} \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u} \right\|_{\mathcal{L}^2}^2 + \int_I \int_{D_{\text{ref}}} |\partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}|^2 \det J \, d\mathbf{x} \, dt \\ & + \int_I \int_{D_{\text{ref}}} A(\mathbf{x}, \mathbf{y}) \nabla \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\mathbf{x}, t, \mathbf{y}) \cdot \nabla \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\mathbf{x}, t, \mathbf{y}) \, d\mathbf{x} \, dt \\ & + \int_{D_{\text{ref}}} |\partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\mathbf{x}, 0, \mathbf{y})|^2 \det J(\mathbf{x}, \mathbf{y}) \, d\mathbf{x} \\ & = \frac{\sigma_{\min}^{2d}}{C_{D_{\text{ref}}}^2} \left\| \frac{\partial}{\partial t} \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u} \right\|_{\mathcal{L}^2}^2 + \int_I \int_{D_{\text{ref}}} A(\mathbf{x}, \mathbf{y}) \nabla \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\mathbf{x}, t, \mathbf{y}) \cdot \nabla \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\mathbf{x}, t, \mathbf{y}) \, d\mathbf{x} \, dt \\ & + \int_{D_{\text{ref}}} |\partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\mathbf{x}, T, \mathbf{y})|^2 \det J(\mathbf{x}, \mathbf{y}) \, d\mathbf{x} \\ & \geq \frac{\sigma_{\min}^{2d}}{C_{D_{\text{ref}}}^2} \left\| \frac{\partial}{\partial t} \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u} \right\|_{\mathcal{L}^2}^2 + \frac{\sigma_{\min}^d}{\sigma_{\max}^2} \left\| \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\cdot, \cdot, \mathbf{y}) \right\|_{L^2(I; H_0^1(D_{\text{ref}}))}^2 \\ & + \sigma_{\min}^d \left\| \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\cdot, T, \mathbf{y}) \right\|_{L^2(D_{\text{ref}})}^2 \\ & \geq \min \left\{ \frac{\sigma_{\min}^{2d}}{C_{D_{\text{ref}}}^2}, \frac{\sigma_{\min}^d}{\sigma_{\max}^2} \right\} \left\| \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\cdot, \cdot, \mathbf{y}) \right\|_{\mathcal{L}}^2. \end{aligned}$$

Step 4. The right-hand side can be bounded from above

$$\begin{aligned} & \sum_{\mathbf{0} \neq \mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} \left[\left\| \frac{\partial}{\partial t} \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \widehat{u}(\cdot, \cdot, \mathbf{y}) \right\|_{\mathcal{L}^2} \left\| \widehat{z}(\cdot, \cdot, \mathbf{y}) \right\|_{\mathcal{L}^2} \sigma_{\max}^d C_{\det J}^{|\mathbf{m}|} \frac{((|\mathbf{m}| + d^2 - 1)!)^\beta}{((d^2 - 1)!)^\beta} \right. \\ & + \left\| \frac{\partial}{\partial t} \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \widehat{u}(\cdot, \cdot, \mathbf{y}) \right\|_{\mathcal{L}^2} \left\| \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\cdot, \cdot, \mathbf{y}) \right\|_{\mathcal{L}^2} \sigma_{\max}^d C_{\det J}^{|\mathbf{m}|} \frac{((|\mathbf{m}| + d^2 - 1)!)^\beta}{((d^2 - 1)!)^\beta} \\ & + \left\| \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \widehat{u}(\cdot, \cdot, \mathbf{y}) \right\|_{L^2(I; H_0^1(D_{\text{ref}}))} \left\| \widehat{z}(\cdot, \cdot, \mathbf{y}) \right\|_{L^2(I; H_0^1(D_{\text{ref}}))} \left. \right] \end{aligned}$$

$$\begin{aligned}
& \times C_{A,1} C_{A,2}^{|\mathbf{m}|} \max\{1, 2^{|\mathbf{m}|-1}\} \frac{((|\mathbf{m}| + d^2)!)^\beta}{(d^2!)^\beta} \\
& + \|\partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \hat{u}(\cdot, \cdot, \mathbf{y})\|_{L^2(I; H_0^1(D_{\text{ref}}))} \|\partial_{\mathbf{y}}^{\mathbf{v}} \hat{u}(\cdot, \cdot, \mathbf{y})\|_{L^2(I; H_0^1(D_{\text{ref}}))} \\
& \times C_{A,1} C_{A,2}^{|\mathbf{m}|} \max\{1, 2^{|\mathbf{m}|-1}\} \frac{((|\mathbf{m}| + d^2)!)^\beta}{(d^2!)^\beta} \\
& + \|\partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \hat{u}(\cdot, 0, \mathbf{y})\|_{L^2(D_{\text{ref}})} \|\partial_{\mathbf{y}}^{\mathbf{v}} \hat{u}(\cdot, 0, \mathbf{y})\|_{L^2(D_{\text{ref}})} \\
& \times \sigma_{\max}^d C_{\det J}^{|\mathbf{m}|} \frac{((|\mathbf{m}| + d^2 - 1)!)^\beta}{((d^2 - 1)!)^\beta} \mathbf{b}^{\mathbf{m}} \\
& + \|\hat{z}(\cdot, \cdot, \mathbf{y})\|_{L^2(I \times D_{\text{ref}})} C_{f_{\text{ref}},1} C_{f_{\text{ref}},2}^{|\mathbf{v}|} \max\{1, \|\boldsymbol{\rho}\|_{\ell^{1/\beta}}^{|\mathbf{v}|}\} \frac{((|\mathbf{v}| + d^2)!)^\beta}{(d^2!)^\beta} \mathbf{b}^{\mathbf{v}} \\
& + \|\partial_{\mathbf{y}}^{\mathbf{v}} \hat{u}(\cdot, \cdot, \mathbf{y})\|_{L^2(I \times D_{\text{ref}})} C_{f_{\text{ref}},1} C_{f_{\text{ref}},2}^{|\mathbf{v}|} \max\{1, \|\boldsymbol{\rho}\|_{\ell^{1/\beta}}^{|\mathbf{v}|}\} \frac{((|\mathbf{v}| + d^2)!)^\beta}{(d^2!)^\beta} \mathbf{b}^{\mathbf{v}} \\
& + \sum_{\mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} \|\partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \hat{u}_0\|_{L^2(D_{\text{ref}})} \|\partial_{\mathbf{y}}^{\mathbf{v}} \hat{u}(\cdot, 0, \mathbf{y})\|_{L^2(D_{\text{ref}})} \\
& \times \sigma_{\max}^d C_{\det J}^{|\mathbf{m}|} \frac{((|\mathbf{m}| + d^2 - 1)!)^\beta}{((d^2 - 1)!)^\beta} \mathbf{b}^{\mathbf{m}}.
\end{aligned}$$

By Evans (2010, Theorem 5.9.3), there holds for some constant $M > 0$

$$\begin{aligned}
\max_{t \in I} \|v(\cdot, t)\|_{L^2(D_{\text{ref}})} & \leq \frac{M}{\sqrt{2}} (\|v\|_{L^2(I; H_0^1(D_{\text{ref}}))} + \|\frac{\partial}{\partial t} v\|_{L^2(I; H^{-1}(D_{\text{ref}}))}) \\
& \leq M \|v\|_{\mathcal{X}},
\end{aligned} \tag{4.9}$$

and there exists $C_{\Delta, \max} > 0$ such that

$$\|(-\Delta)^{-1} v\|_{H_0^1(D_{\text{ref}}) \cap H^2(D_{\text{ref}})} \leq \frac{C_{\Delta, \max}}{M \sigma_{\max}^d \sqrt{T}} \|v\|_{L^2(D_{\text{ref}})}$$

for all $v \in L^2(D_{\text{ref}})$. In particular, we have that

$$\|\hat{z}\|_{L^2(I \times D_{\text{ref}})} \leq \frac{C_{\Delta, \max}}{M \sigma_{\max}^d \sqrt{T}} \|\det J \partial_{\mathbf{y}}^{\mathbf{v}} \partial_t \hat{u}\|_{\mathcal{L}^2} \leq C_{\Delta, \max} \|\partial_{\mathbf{y}}^{\mathbf{v}} \hat{u}\|_{\mathcal{X}}. \tag{4.10}$$

Moreover, since we have that

$$-\Delta \hat{z} = \det J \partial_{\mathbf{y}}^{\mathbf{v}} \hat{u} \quad \text{in } L^2(D_{\text{ref}}),$$

we infer

$$\int_{D_{\text{ref}}} \nabla \hat{z} \cdot \nabla \hat{v} \, d\mathbf{x} = \int_{D_{\text{ref}}} (\partial_{\mathbf{y}}^{\mathbf{v}} \hat{u}) \hat{v} \det J \, d\mathbf{x} \quad \text{for all } \hat{v} \in H_0^1(D_{\text{ref}}).$$

In consequence,

$$\|\nabla \hat{z}\|_{L^2(D_{\text{ref}})}^2 \leq \sigma_{\max}^d \|\partial_{\mathbf{y}}^{\mathbf{v}} \hat{u}\|_{L^2(D_{\text{ref}})} \|\hat{z}\|_{L^2(D_{\text{ref}})}$$

$$\leq \sigma_{\max}^d C_{D_{\text{ref}}} \|\partial_{\mathbf{y}}^{\mathbf{v}} \hat{u}\|_{L^2(D_{\text{ref}})} \|\nabla \hat{z}\|_{L^2(D_{\text{ref}})},$$

whence

$$\|\nabla \hat{z}\|_{L^2(D_{\text{ref}})} \leq \sigma_{\max}^d C_{D_{\text{ref}}} \|\partial_{\mathbf{y}}^{\mathbf{v}} \hat{u}\|_{L^2(D_{\text{ref}})}. \quad (4.11)$$

Hence, the right-hand side has the upper bound

$$\begin{aligned} & \sum_{\mathbf{0} \neq \mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} \|\partial_{\mathbf{y}}^{\mathbf{v}} \hat{u}(\cdot, \cdot, \mathbf{y})\|_{\mathcal{X}} \mathbf{b}^{\mathbf{m}} \\ & \times \left[\|\partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \hat{u}(\cdot, \cdot, \mathbf{y})\|_{\mathcal{X}} C_{\Delta, \max} \sigma_{\max}^{2d} C_{\det J}^{\|\mathbf{m}\|} \frac{((\|\mathbf{m}\| + d^2 - 1)!)^\beta}{((d^2 - 1)!)^\beta} \right. \\ & + \|\partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \hat{u}(\cdot, \cdot, \mathbf{y})\|_{\mathcal{X}} C_{D_{\text{ref}}} \sigma_{\max}^d C_{\det J}^{\|\mathbf{m}\|} \frac{((\|\mathbf{m}\| + d^2 - 1)!)^\beta}{((d^2 - 1)!)^\beta} \\ & + \sigma_{\max}^d C_{D_{\text{ref}}}^2 \|\partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \hat{u}(\cdot, \cdot, \mathbf{y})\|_{\mathcal{X}} C_{A,1} C_{A,2}^{\|\mathbf{m}\|} \max\{1, 2^{\|\mathbf{m}\|-1}\} \frac{((\|\mathbf{m}\| + d^2)!)^\beta}{(d^2!)^\beta} \\ & + \|\partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \hat{u}(\cdot, \cdot, \mathbf{y})\|_{\mathcal{X}} C_{A,1} C_{A,2}^{\|\mathbf{m}\|} \max\{1, 2^{\|\mathbf{m}\|-1}\} \frac{((\|\mathbf{m}\| + d^2)!)^\beta}{(d^2!)^\beta} \\ & + M^2 \|\partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \hat{u}(\cdot, \cdot, \mathbf{y})\|_{\mathcal{X}} \sigma_{\max}^d C_{\det J}^{\|\mathbf{m}\|} \frac{((\|\mathbf{m}\| + d^2 - 1)!)^\beta}{((d^2 - 1)!)^\beta} \Big] \\ & + \|\partial_{\mathbf{y}}^{\mathbf{v}} \hat{u}(\cdot, \cdot, \mathbf{y})\|_{\mathcal{X}} C_{\Delta, \max} \sigma_{\max}^d C_{f_{\text{ref}},1} C_{f_{\text{ref}},2}^{\|\mathbf{v}\|} \max\{1, \|\boldsymbol{\rho}\|_{\ell^{1/\beta}}^{\|\mathbf{v}\|}\} \frac{((\|\mathbf{v}\| + d^2)!)^\beta}{(d^2!)^\beta} \mathbf{b}^{\mathbf{v}} \\ & + C_{D_{\text{ref}}} \|\partial_{\mathbf{y}}^{\mathbf{v}} \hat{u}(\cdot, \cdot, \mathbf{y})\|_{\mathcal{X}} C_{f_{\text{ref}},1} C_{f_{\text{ref}},2}^{\|\mathbf{v}\|} \max\{1, \|\boldsymbol{\rho}\|_{\ell^{1/\beta}}^{\|\mathbf{v}\|}\} \frac{((\|\mathbf{v}\| + d^2)!)^\beta}{(d^2!)^\beta} \mathbf{b}^{\mathbf{v}} \\ & + \sum_{\mathbf{m} \leq \mathbf{v}} \left[\binom{\mathbf{v}}{\mathbf{m}} M \|\partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \hat{u}_0\|_{L^2(D_{\text{ref}})} \|\partial_{\mathbf{y}}^{\mathbf{v}} \hat{u}(\cdot, \cdot, \mathbf{y})\|_{\mathcal{X}} \sigma_{\max}^d C_{\det J}^{\|\mathbf{m}\|} \right. \\ & \quad \left. \times \frac{((\|\mathbf{m}\| + d^2 - 1)!)^\beta}{((d^2 - 1)!)^\beta} \mathbf{b}^{\mathbf{m}} \right], \end{aligned}$$

where we applied the inequalities (4.9)–(4.11) as well as the Poincaré inequality multiple times.

Step 5. We derive the following bound on the pullback initial condition.

Lemma 12. *Under Assumptions (A1)–(A3) and (A8), there holds*

$$\|\partial_{\mathbf{y}}^{\mathbf{v}} \hat{u}_0(\cdot, \mathbf{y})\|_{L^2(D_{\text{ref}})} \leq 2^{\beta\|\mathbf{v}\|-\beta} d^{\beta\|\mathbf{v}\|} C_{u_0} C^{\|\mathbf{v}\|} \mathbf{b}^{\mathbf{v}} (|\mathbf{v}|!)^\beta$$

for all $\mathbf{v} \in \mathcal{F} \setminus \{\mathbf{0}\}$ and $\mathbf{y} \in U$.

Proof. We use Faà di Bruno's formula (Savits, 2006) to obtain

$$\partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}_0(\mathbf{x}, \mathbf{y}) = \sum_{\substack{\boldsymbol{\lambda} \in \mathbb{N}_0^d \\ 1 \leq |\boldsymbol{\lambda}| \leq |\mathbf{v}|}} \partial_{\mathbf{x}}^{\boldsymbol{\lambda}} u_0(\mathbf{x}) \Big|_{\mathbf{x}=\mathbf{V}(\mathbf{x}, \mathbf{y})} \alpha_{\mathbf{v}, \boldsymbol{\lambda}}(\mathbf{x}, \mathbf{y}),$$

where the coefficients $(\alpha_{\mathbf{v}, \boldsymbol{\lambda}})$ are defined exactly as in Lemma 9. In particular,

$$\|\alpha_{\mathbf{v}, \boldsymbol{\lambda}}(\cdot, \mathbf{y})\|_{L^\infty} \leq C^{|\mathbf{v}|} \frac{(|\mathbf{v}|!)^\beta (|\mathbf{v}| - 1!)^\beta}{(\boldsymbol{\lambda}!)^\beta (|\mathbf{v}| - |\boldsymbol{\lambda}|!)^\beta (|\boldsymbol{\lambda}| - 1!)^\beta} \mathbf{b}^{\mathbf{v}}.$$

Together with Assumption (A8), this yields

$$\|\partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}_0(\cdot, \mathbf{y})\|_{L^2(D_{\text{ref}})} \leq C_{u_0} C^{|\mathbf{v}|} \mathbf{b}^{\mathbf{v}} \sum_{\substack{\boldsymbol{\lambda} \in \mathbb{N}_0^d \\ 1 \leq |\boldsymbol{\lambda}| \leq |\mathbf{v}|}} \frac{(|\mathbf{v}|!)^\beta (|\mathbf{v}| - 1!)^\beta}{(\boldsymbol{\lambda}!)^\beta (|\mathbf{v}| - |\boldsymbol{\lambda}|!)^\beta (|\boldsymbol{\lambda}| - 1!)^\beta}.$$

The claim follows in complete analogy to the proof of Lemma 9 in the special case $\boldsymbol{\rho} = \mathbf{1} \in \mathbb{R}^d$. $\square$

Step 6. Putting the lower bound together with the upper bound yields the recurrence relation

$$\begin{aligned} \|\widehat{u}(\cdot, \cdot, \mathbf{y})\|_{\mathcal{X}} &\leq C_0 \\ \|\partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\cdot, \cdot, \mathbf{y})\|_{\mathcal{X}} &\leq \widetilde{C}_1 \sum_{\mathbf{0} \neq \mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} \widetilde{C}_2^{|\mathbf{m}|} \|\partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \widehat{u}(\cdot, \cdot, \mathbf{y})\|_{\mathcal{X}} (|\mathbf{m}| + d^2)^\beta \mathbf{b}^{\mathbf{m}} \\ &\quad + \widetilde{C}_1 \widetilde{C}_2^{|\mathbf{v}|} (|\mathbf{v}| + d^2)^\beta \mathbf{b}^{\mathbf{v}} \\ &\quad + \widetilde{C}_1 \sum_{\mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} \|\partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \widehat{u}_0\|_{L^2(D_{\text{ref}})} \widetilde{C}_2^{|\mathbf{m}|} (|\mathbf{m}| + d^2)^\beta \mathbf{b}^{\mathbf{m}}, \end{aligned}$$

for all $\mathbf{v} \in \mathcal{F} \setminus \{\mathbf{0}\}$, where

$$\begin{aligned} C_0 &= \widetilde{C}_1 (1 + \|\widehat{u}_0\|_{L^2}), \\ \widetilde{C}_1 &= (C_{\Delta, \max} \sigma_{\max}^{2d} + C_{D_{\text{ref}}} \sigma_{\max}^d + \sigma_{\max}^d C_{D_{\text{ref}}}^2 C_{A,1} + C_{A,1} + M^2 \sigma_{\max}^d \\ &\quad + C_{\Delta, \max} \sigma_{\max}^d C_{f_{\text{ref}},1} + C_{D_{\text{ref}}} C_{f_{\text{ref}},1} + M \sigma_{\max}^d) \min \left\{ \frac{\sigma_{\min}^{2d}}{C_{D_{\text{ref}}}^2 C_{\Delta}^2}, \frac{\sigma_{\min}^d}{\sigma_{\max}^2} \right\}^{-1}, \\ \widetilde{C}_2 &= 4C_{\det J} + 4C_{A,2} + 2C_{f_{\text{ref}},2} \max \{1, \|\boldsymbol{\rho}\|_{\ell^{1/\beta}}\}. \end{aligned}$$

By Lemma 12, we can simplify the third term in the recurrence bound as follows:

$$\begin{aligned} &\sum_{\mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} \|\partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \widehat{u}_0\|_{L^2(D_{\text{ref}})} \widetilde{C}_2^{|\mathbf{m}|} (|\mathbf{m}| + d^2)^\beta \mathbf{b}^{\mathbf{m}} \\ &\leq 2^{\beta|\mathbf{v}|-\beta} d^{\beta|\mathbf{v}|} C_{u_0} C^{|\mathbf{v}|} \widetilde{C}_2^{|\mathbf{v}|} \mathbf{b}^{\mathbf{v}} \sum_{\mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} (|\mathbf{v}| - |\mathbf{m}|)^\beta (|\mathbf{m}| + d^2)^\beta. \end{aligned}$$

Here, we can further simplify

$$\begin{aligned}
& \sum_{\mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} ((|\mathbf{v}| - |\mathbf{m}|)!)^\beta ((|\mathbf{m}| + d^2)!)^\beta \\
&= \sum_{\ell=0}^{|\mathbf{v}|} ((|\mathbf{v}| - \ell)!)^\beta ((\ell + d^2)!)^\beta \sum_{\substack{|\mathbf{m}|=\ell \\ \mathbf{m} \leq \mathbf{v}}} \binom{\mathbf{v}}{\mathbf{m}} \\
&= \sum_{\ell=0}^{|\mathbf{v}|} ((|\mathbf{v}| - \ell)!)^\beta ((\ell + d^2)!)^\beta \binom{|\mathbf{v}|}{\ell} \\
&\leq \sum_{\ell=0}^{|\mathbf{v}|} ((|\mathbf{v}| - \ell)!)^\beta ((\ell + d^2)!)^\beta \binom{|\mathbf{v}|}{\ell}^\beta \\
&= (|\mathbf{v}|!)^\beta \sum_{\ell=0}^{|\mathbf{v}|} \frac{((\ell + d^2)!)^\beta}{(\ell!)^\beta} \\
&\leq (|\mathbf{v}|!)^\beta \left(\sum_{\ell=0}^{|\mathbf{v}|} \frac{(\ell + d^2)!}{\ell!} \right)^\beta \\
&= (|\mathbf{v}|!)^\beta \left(\frac{(|\mathbf{v}| + d^2 + 1)!}{(d^2 + 1)! |\mathbf{v}|!} \right)^\beta = \frac{((|\mathbf{v}| + d^2 + 1)!)^\beta}{((d^2 + 1)!)^\beta}.
\end{aligned}$$

This means that the recurrence can be expressed as

$$\begin{aligned}
& \|\widehat{u}(\cdot, \cdot, \mathbf{y})\|_{\mathcal{X}} \leq C_0 \\
& \|\partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\cdot, \cdot, \mathbf{y})\|_{\mathcal{X}} \leq C_1 \sum_{\mathbf{0} \neq \mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} C_2^{|\mathbf{m}|} \|\partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \widehat{u}(\cdot, \cdot, \mathbf{y})\|_{\mathcal{X}} ((|\mathbf{m}| + d^2)!)^\beta \mathbf{b}^{\mathbf{m}} \\
& \quad + C_1 C_2^{|\mathbf{v}|} ((|\mathbf{v}| + d^2 + 1)!)^\beta \mathbf{b}^{\mathbf{v}},
\end{aligned}$$

with $C_1 = \widetilde{C}_1 + C_{u_0}$ and $C_2 = \widetilde{C}_2 + 2^\beta d^\beta C \widetilde{C}_2$. We can now apply Lemma 21 to obtain the inductive bound.

Theorem 13. *Let $\mathbf{v} \in \mathcal{F}$ and $\mathbf{y} \in U$. Under Assumptions (A1)–(A4) as well as Assumptions (A8)–(A10), the solution to (2.3) satisfies*

$$\|\partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\cdot, \cdot, \mathbf{y})\|_{\mathcal{X}} \leq C_{\widehat{u},1} C_{\widehat{u},2}^{|\mathbf{v}|} (|\mathbf{v}|!)^\beta \mathbf{b}^{\mathbf{v}},$$

where the space $\mathcal{X}$ is defined in (2.4) and

$$\begin{aligned}
C_{\widehat{u},1} &= 1 + C_0, \\
C_{\widehat{u},2} &= \widetilde{C}^2 ((d^2 + 1)!)^\beta 2^{\beta(d^2+2)+1}, \\
\widetilde{C} &= \max\{\widetilde{C}_1 + C_{u_0}, \widetilde{C}_2 + 2^\beta d^\beta C \widetilde{C}_2\}, \\
C_0 &= \widetilde{C}_1 (1 + C_{u_0}).
\end{aligned}$$

5. Error analysis

For the error analysis of the numerical treatment of the considered problems (2.2) and (2.3) one needs to take three types of errors into account. The first one comes from the dimension truncation of the input random field to finitely many terms. The second error is caused by QMC cubature and the last one is the finite element discretization error. Combining these estimates will yield an overall error estimate. Throughout this section the mentioned errors will be stated for both, the stationary and the parabolic problem. We will measure the error for the stationary case in $\|\cdot\|_{H_0^1(D_{\text{ref}})}$ and in the parabolic problem the error will be measured in the norm $\|\cdot\|_{L^2(I;H_0^1(D_{\text{ref}}))}$.

5.1. Truncation error

In the case of the dimension truncation error one wants to find an upper bound of the expression

$$\left\| \int_U (\hat{u}(\mathbf{y}) - \hat{u}_s(\mathbf{y})) d\mathbf{y} \right\|_Z,$$

where $Z = H_0^1(D_{\text{ref}})$ if $\hat{u}$ is the solution of the stationary problem (2.2) and $Z = L^2(I;H_0^1(D_{\text{ref}}))$ if we consider $\hat{u}$ to be the solution to the space-time formulation of the parabolic equation (2.3). Here $\hat{u}_s$ denotes the dimensionally truncated PDE solution, which is defined as

$$\hat{u}_s(\mathbf{y}) := \hat{u}(y_1, \dots, y_s, 0, 0, \dots) \text{ for } \mathbf{y} \in U, \quad s \in \mathbb{N}.$$

The dimension truncation rate follows as an immediate consequence of Theorems 11 and 13 together with Guth and Kaarnioja (2024, Theorem 4.3).

Theorem 14. *Under Assumptions (A1)–(A10), there exists a constant $C_{\text{dim}} > 0$ independent of s such that*

$$\left\| \int_U (\hat{u}(\mathbf{y}) - \hat{u}_s(\mathbf{y})) d\mathbf{y} \right\|_Z \leq C_{\text{dim}} s^{-\frac{2}{p}+1}, \quad (5.1)$$

where $p \in (0, 1)$ as in Assumption (A5) and we set $Z = H_0^1(D_{\text{ref}})$ if $\hat{u}$ is the solution to the pullback Poisson equation. If $\hat{u}$ is the solution to the pullback heat equation, then under the additional Assumption (A8) inequality (5.1) holds with $Z = L^2(I;H_0^1(D_{\text{ref}}))$.

5.2. QMC error

We want to quantify the error resulting from the approximation of an integral over the s -dimensional unit cube using QMC cubature, which was introduced in Section 3.

The following theorem gives a choice of POD weights as well as a QMC error bound in the sense of the root-mean square error independent of the dimension s .

Theorem 15. *Let $\hat{u}_s$ denote either the solution to the Poisson problem, in which case we assume that $Z = H_0^1(D_{\text{ref}})$ and Assumptions (A1)–(A5) hold, or let $\hat{u}_s$ denote the solution to the heat equation, in which case we assume that $Z = L^2(I;H_0^1(D_{\text{ref}}))$ and Assumptions (A1)–(A5) and (A8)–(A10). Then the root-mean-square error using a randomly shifted lattice rule (3.1) with $n = 2^m$, $m \in \mathbb{N}$, points constructed by a CBC algorithm with R independent random shifts satisfies*

$$\sqrt{\mathbb{E}_{\mathbf{A}} \|I_s(\hat{u}_s) - Q_{\text{ran}}(\hat{u}_s)\|_Z^2} \leq C \cdot n^{-\min\{\frac{1}{p}-\frac{1}{2}, 1-\varepsilon\}} \quad \text{for arbitrary } \varepsilon \in (0, \frac{1}{2}),$$

where $C > 0$ is independent of s when the weights $(\gamma_u)_{u \in \mathbb{N}}$ are chosen as the sequence of product and order dependent (POD) weights

$$\gamma_u = \left((|u|!)^\beta \prod_{j \in u} \frac{C_{\hat{u},2} b_j}{\sqrt{2\zeta(2\lambda)/(2\pi^2)^\lambda}} \right)^{\frac{2}{1+\lambda}},$$

$$\lambda = \begin{cases} \frac{p}{2-p} & \text{if } p \in (\frac{2}{3}, \frac{1}{\beta}), \\ \frac{1}{2-2\varepsilon} & \text{if } p \in (0, \min\{\frac{2}{3}, \frac{1}{\beta}\}], p \neq \frac{1}{\beta}, \end{cases}$$

where $C_{\hat{u},2}$ is defined in Theorems 11–13.

Proof. The proof is carried out in complete analogy to Kuo et al. (2012). $\square$

5.3. Finite element error

Let Assumptions (A1)–(A5) and (A7) hold. For the finite element discretization with respect to the spatial variable $\mathbf{x} \in D_{\text{ref}}$, we consider finite element subspaces of $H_0^1(D_{\text{ref}})$, denoted by $\{V_h\}_h$, where h refers to the mesh size of a regular triangulation of D_{ref} . As the basis functions we choose continuous piecewise linear functions spanning the finite element spaces. With respect to the temporal variable $t \in I$ in the pullback heat equation, we shall employ an implicit Euler time discretization with time step size $\Delta t > 0$. For further details, we refer to Ern and Guermond (2021).

Theorem 16.

- (i) Let $\hat{u}_{s,h}(\cdot, \mathbf{y})$, $\mathbf{y} \in U$, denote the dimensionally truncated finite element solution to the pullback Poisson equation with finite element mesh width $h > 0$. Under Assumptions (A1)–(A5) and (A7), there holds

$$\|\hat{u}_s(\cdot, \mathbf{y}) - \hat{u}_{s,h}(\cdot, \mathbf{y})\|_{L^2(D_{\text{ref}})} \leq C_{\text{FEM}} h^2,$$

where the constant $C_{\text{FEM}} > 0$ is independent of s , h , and $\mathbf{y}$.

- (ii) Let $\hat{u}_{s,h,\Delta t}(\cdot, \cdot, \mathbf{y})$, $\mathbf{y} \in U$, denote the dimensionally truncated space-time discretized solution to the pullback heat equation obtained using an implicit Euler time discretization with time step size $\Delta t > 0$ and finite element mesh width $h > 0$. Under Assumptions (A1)–(A5) and (A7)–(A10), there holds

$$\|\hat{u}_s(\cdot, \cdot, \mathbf{y}) - \hat{u}_{s,h,\Delta t}(\cdot, \cdot, \mathbf{y})\|_{L^2(I; L^2(D_{\text{ref}}))} \leq C_{\text{FEM}} (\Delta t + h^2),$$

where the constant $C_{\text{FEM}} > 0$ is independent of s , h , Δt , and $\mathbf{y}$.

Proof. In the stationary case, we refer to Hakula et al. (2024). For the time-dependent case, we refer the reader to Meidner and Vexler (2008). $\square$

5.4. Overall error

The combination of the aforementioned error estimates, i.e., truncation error, QMC error and finite element error yields an overall error bound for both considered problems.

Theorem 17.

- (i) Let $\hat{u}$ be the solution of the pullback variational formulation of the Poisson equation (2.2) and let $\hat{u}_{s,h}$ denote the dimensionally truncated finite element solution. Under the Assumptions (A1)–(A7) we find in the stationary setting

$$\sqrt{\mathbb{E}_\Delta \|I_s(\hat{u}) - Q_{\text{ran}}(\hat{u}_{s,h})\|_{L^2(D_{\text{ref}})}^2} \leq C_{\text{all}} \left(s^{-\frac{2}{p}+1} + h^2 + n^{-\min\{\frac{1}{p}-\frac{1}{2}, 1-\varepsilon\}} \right)$$

with s denoting the truncation index, h being the mesh size used in the finite element discretization and $C_{\text{all}} > 0$ being a positive constant, which is independent of s, h and the number of lattice points n .

- (ii) Let $\hat{u}$ denote the solution to the pullback weak formulation of the heat equation (2.3) and let $\hat{u}_{s,h,\Delta t}$ denote the dimensionally truncated space-time finite element solution. For the parabolic case one obtains under Assumptions (A1)–(A10) the overall error bound

$$\begin{aligned} & \sqrt{\mathbb{E}_\Delta \|I_s(\hat{u}) - Q_{\text{ran}}(\hat{u}_{s,h,\Delta t})\|_{L^2(I; L^2(D_{\text{ref}}))}^2} \\ & \leq C_{\text{all}} \left(s^{-\frac{2}{p}+1} + h^2 + n^{-\min\{\frac{1}{p}-\frac{1}{2}, 1-\varepsilon\}} + \Delta t \right), \end{aligned}$$

where Δt denotes the step size of the implicit Euler scheme for the discretization of the temporal variable and $C_{\text{all}} > 0$ is a constant, which does not depend on s, h, n and Δt .

Proof. The claim follows by combining Theorems 14–16. $\square$

6. Numerical Experiments

In this section, we consider a series of numerical experiments to approximate the expected values of solutions to the Poisson equation and the heat equation subject to domain uncertainty using QMC. For the numerical experiments, we fix the reference domain

$$D_{\text{ref}} = \{\mathbf{x} \in \mathbb{R}^2 \mid x_1^2 + x_2^2 \leq 1\} \subset \mathbb{R}^2$$

and consider domains parameterized by $D(\mathbf{y}) = \mathbf{V}(D_{\text{ref}}, \mathbf{y})$ for $\mathbf{y} \in U_s$, where

$$\begin{aligned} \mathbf{V}(\mathbf{x}, \mathbf{y}) &= \left[1 + \xi \left(\sum_{j=1}^s y_j \psi_j(\mathbf{x}) \right) \right] \mathbf{x}, \\ \psi_j(\mathbf{x}) &= j^{-\vartheta} \sin(3j(\text{atan2}(x_1, x_2) + \pi)), \quad j \geq 1, \end{aligned}$$

for $\mathbf{x} \in D_{\text{ref}}$ and $\mathbf{y} \in U_s$. We set $s = 100$ as the truncation dimension and consider the following four experiments:

- (E1) $\xi(y) = y$ and $\vartheta = 2.1$;
- (E2) $\xi(y) = \exp\left(-\frac{1}{y+\xi(\vartheta)/2}\right)$ and $\vartheta = 2.1$;
- (E3) $\xi(y) = y$ and $\vartheta = 2.5$;
- (E4) $\xi(y) = \exp\left(-\frac{1}{y+\xi(\vartheta)/2}\right)$ and $\vartheta = 2.5$.

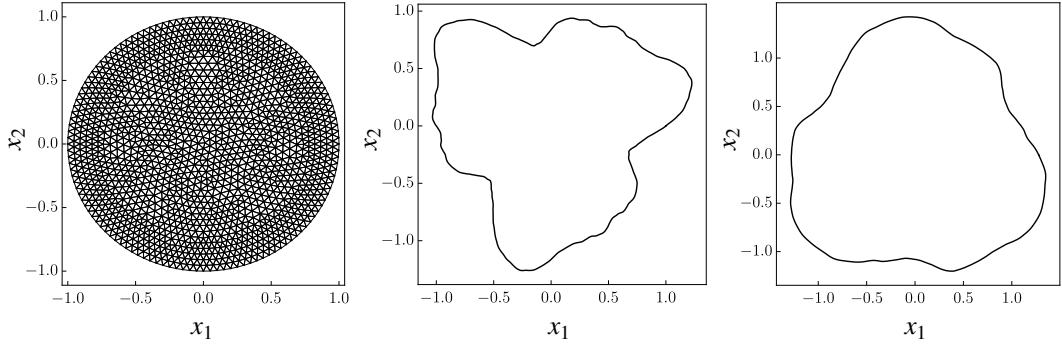


FIG. 1. Left: the triangulation of the reference domain D_{ref} used in all numerical experiments. Middle: a random realization of the random field corresponding to experiment (E1). Right: a random realization of the random field corresponding to experiment (E2).

Experiments (E1) and (E3) correspond to vectorized Karhunen–Loève expansions and are thereby covered not only by our theory, but also by the existing results presented in Harbrecht et al. (2016). Meanwhile, experiments (E2) and (E4) do not produce domain mappings which are holomorphic with respect to the parametric variable $\mathbf{y}$, but they are covered by the theory developed for Gevrey regular domain mappings in the present work. Some realizations of the parametric domains corresponding to experiments (E1) and (E2) are illustrated in Figure 1 for randomly generated $\mathbf{y} \in U_s$ alongside the finite element discretization used for the reference domain D_{ref} . The same finite element mesh is used for all experiments. In particular, we solve the pullback weak formulation of the Poisson equation and the heat equation using a first-order finite element mesh with mesh width $h = 0.1$. Meanwhile, the pullback heat equation is solved by first discretizing the temporal variable using the implicit Euler method with time step size $\Delta t = 0.1$. The heat equation was solved over the time interval $I = [0, 1]$ subject to the initial heat distribution $u_0 \equiv 1$.

It is straightforward to verify that

$$\|\partial_{\mathbf{y}}^{\mathbf{v}} \mathbf{V}(\cdot; \mathbf{y})\|_{W^{1,\infty}(D_{\text{ref}})} \lesssim \begin{cases} |\mathbf{v}|! \mathbf{b}^{\mathbf{v}} & \text{in experiments (E1) and (E3),} \\ (|\mathbf{v}|!)^2 \mathbf{b}^{\mathbf{v}} & \text{in experiments (E2) and (E4),} \end{cases}$$

where $\mathbf{b} = (b_j)_{j \geq 1}$ is defined by $b_j = \|\psi_j\|_{W^{1,\infty}(D_{\text{ref}})} \propto j^{1-\vartheta}$. For numerical stability, we set $C_{\hat{u},2} = 1$ in the expression for the QMC weights derived in Theorem 15 and use the QMC4PDE software (Kuo and Nuyens, 2016b) to obtain the generating vector for $\vartheta \in \{2.1, 2.5\}$. For the approximation of $\mathbb{E}[\hat{u}]$, we use $R = 16$ random shifts. The root-mean-square (R.M.S.) errors are approximated using the formula

$$\sqrt{\frac{1}{R(R-1)} \sum_{r=1}^R \|Q_{\text{ran}}(\hat{u}) - Q_r(\hat{u})\|_{\mathcal{H}}^2},$$

where we used both $\mathcal{H} = H_0^1(D_{\text{ref}})$ and $\mathcal{H} = L^2(D_{\text{ref}})$ for the solution to the Poisson equation and $\mathcal{H} = \mathcal{L}^2$ and $\mathcal{H} = L^2(I; H_0^1(D_{\text{ref}}))$ for the heat equation. Since the Poincaré inequality can be used to bound $\|\cdot\|_{L^2}$ and $\|\cdot\|_{\mathcal{L}^2}$ by $\|\cdot\|_{H_0^1}$ and $\|\cdot\|_{L^2(I; H_0^1(D_{\text{ref}}))}$, respectively, we expect the same rates of QMC convergence in all experiments. The calculated R.M.S. errors are displayed in Figures 2–3 for the Poisson problem and in Figures 4–5 for the heat equation corresponding to the two different norms.

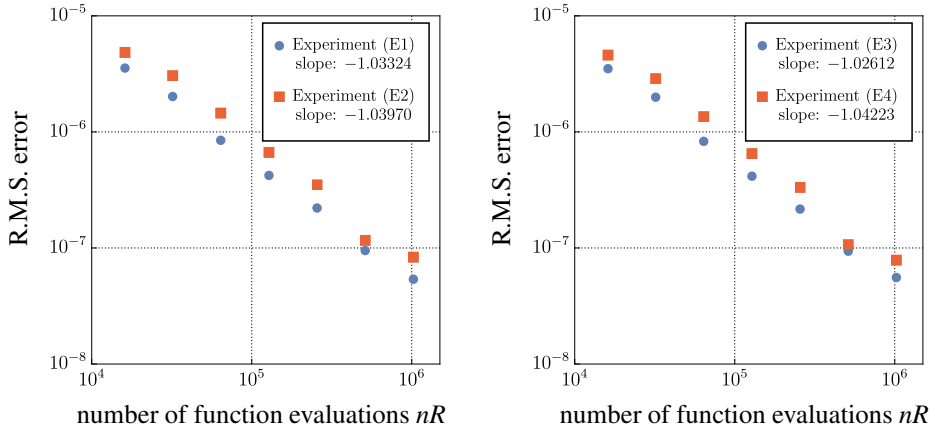


FIG. 2. The computed $L^2(D_{\text{ref}})$ errors for the Poisson equation. Left: the numerical cubature errors corresponding to experiments (E1)–(E2). Right: the numerical cubature errors corresponding to experiments (E3)–(E4).

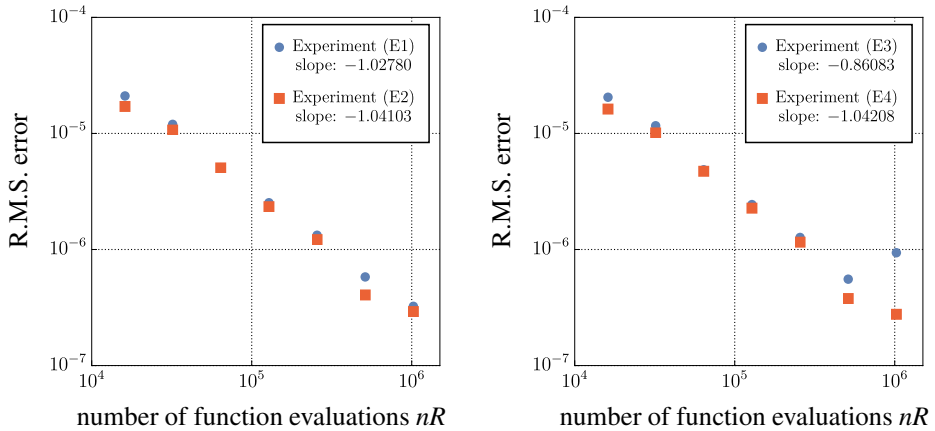


FIG. 3. The computed $H_0^1(D_{\text{ref}})$ errors for the Poisson equation. Left: the numerical cubature errors corresponding to experiments (E1)–(E2). Right: the numerical cubature errors corresponding to experiments (E3)–(E4).

In all cases, we observe that the R.M.S. error decays at an essentially linear rate $\mathcal{O}(n^{-1})$, as proved in Theorem 15. However, since the L^2 and $\mathcal{L}^2$ norms are weaker, the magnitudes of the computed errors are smaller than the errors measured using Sobolev norms. In addition, we note that the H_0^1 norm appears to be sensitive to numerical discretization errors, resulting in a loss of accuracy in the R.M.S. error below a certain threshold as displayed in Figure 3.

7. Conclusions

In this work, we have investigated uncertainty quantification for random domains subject to the Poisson equation and the heat equation. Specifically, we find that modeling the input random fields as Gevrey regular vector fields leads to faster-than-Monte Carlo convergence rates for the computation

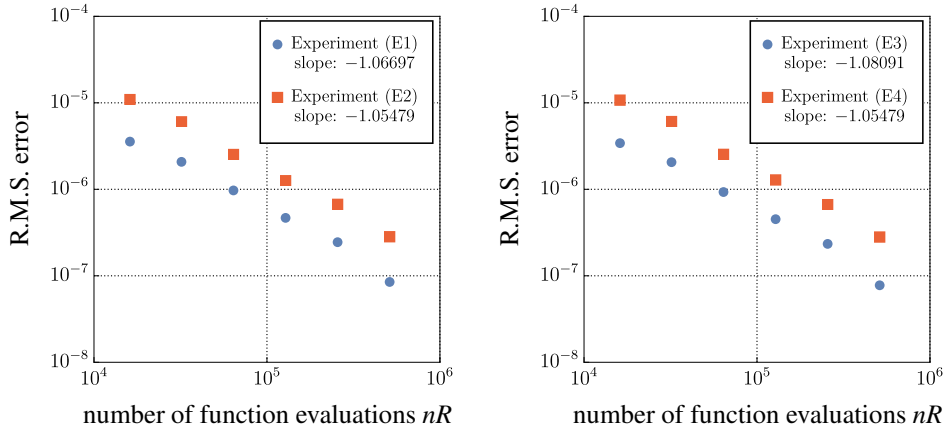


FIG. 4. The computed $L^2(I; L^2(D_{\text{ref}}))$ errors for the heat equation. Left: the numerical cubature errors corresponding to experiments (E1)–(E2). Right: the numerical cubature errors corresponding to experiments (E3)–(E4).

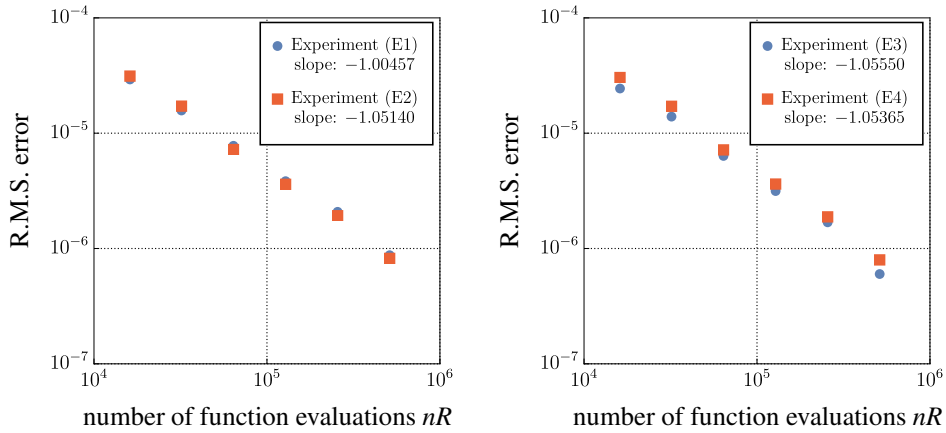


FIG. 5. The computed $L^2(I; H_0^1(D_{\text{ref}}))$ errors for the heat equation. Left: the numerical cubature errors corresponding to experiments (E1)–(E2). Right: the numerical cubature errors corresponding to experiments (E3)–(E4).

of the statistical response when using randomly shifted rank-1 lattice rules. The Gevrey model accommodates more general parameterizations of domain uncertainty: while the previous literature is mainly concerned with holomorphic transformations of a fixed reference domain, the Gevrey class also covers non-holomorphic representations.

For the construction of the QMC rules, we performed a detailed parametric regularity analysis informing the choice of optimized QMC point sets. In addition to the numerical cubature error, we also analyzed the approximation errors stemming from dimension truncation and finite element discretization. We remark that our QMC analysis is carried out in a general manner and thus independent of the chosen PDE discretization scheme. Some potential future directions include extending QMC analysis for curved or time dependent domains as well as forward and inverse domain uncertainty quantification governed by more involved elliptic or parabolic PDEs.

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Appendix

This section covers results on technical recurrence relations and PDE regularity used in our analysis.

Let

$$\tau_{0,\beta,q} = 1 \quad \text{and} \quad \tau_{k,\beta,q} = \sum_{\ell=0}^{k-1} \frac{((k-\ell+q)!)^\beta}{((k-\ell)!)^\beta} \tau_{\ell,\beta,q}, \quad k \geq 1. \quad (7.1)$$

This sequence is a generalization of sequence A003480 in *The On-Line Encyclopedia of Integer Sequences (OEIS)*.

The sequence (7.1) appears in the parametric regularity analysis developed in Section 2 via the following abstract recursive bound.

Lemma 18. *Let $(\Upsilon_{\mathbf{v}})_{\mathbf{v} \in \mathcal{F}}$ be a sequence defined by the recurrence relation*

$$\Upsilon_{\mathbf{0}} = 1 \quad \text{and} \quad \Upsilon_{\mathbf{m}} = \sum_{\substack{\mathbf{w} \leq \mathbf{m} \\ \mathbf{w} \neq \mathbf{0}}} \binom{\mathbf{m}}{\mathbf{w}} ((|\mathbf{w}| + q)!)^\beta \Upsilon_{\mathbf{m}-\mathbf{w}}, \quad \mathbf{m} \in \mathcal{F} \setminus \{\mathbf{0}\}.$$

Then there holds

$$\Upsilon_{\mathbf{v}} \leq \tau_{|\mathbf{v}|,\beta,q} (|\mathbf{v}|!)^\beta \quad \text{for all } \mathbf{v} \in \mathcal{F},$$

where the sequence $(\tau_{k,\beta,q})_{k \geq 0}$ is defined by (7.1).

Proof. We prove the claim by induction with respect to the order of the multi-index $\mathbf{v}$. The claim is clearly true if $\mathbf{v} = \mathbf{0}$. We fix $\mathbf{v} \in \mathcal{F} \setminus \{\mathbf{0}\}$ and assume that the claim holds for all multi-indices with order less than $|\mathbf{v}|$. Then there holds

$$\begin{aligned} \Upsilon_{\mathbf{v}} &= \sum_{\substack{\mathbf{w} \leq \mathbf{v} \\ \mathbf{w} \neq \mathbf{0}}} \binom{\mathbf{v}}{\mathbf{w}} ((|\mathbf{w}| + q)!)^\beta \Upsilon_{\mathbf{v}-\mathbf{w}} \\ &= \sum_{\substack{\mathbf{w} \leq \mathbf{v} \\ \mathbf{w} \neq \mathbf{v}}} \binom{\mathbf{v}}{\mathbf{w}} ((|\mathbf{v}| - |\mathbf{w}| + q)!)^\beta \Upsilon_{\mathbf{w}} \\ &\leq \sum_{\substack{\mathbf{w} \leq \mathbf{v} \\ \mathbf{w} \neq \mathbf{v}}} \binom{\mathbf{v}}{\mathbf{w}} ((|\mathbf{v}| - |\mathbf{w}| + q)!)^\beta \tau_{|\mathbf{w}|,\beta,q} (|\mathbf{w}|!)^\beta \end{aligned}$$

$$\begin{aligned}
&= \sum_{\ell=0}^{|\mathbf{v}|-1} ((|\mathbf{v}| - \ell + q)!)^\beta \tau_{\ell,\beta,q}(\ell!)^\beta \sum_{\substack{\mathbf{w} \leq \mathbf{v} \\ |\mathbf{w}|=\ell}} \binom{\mathbf{v}}{\mathbf{w}} \\
&\leq \sum_{\ell=0}^{|\mathbf{v}|-1} ((|\mathbf{v}| - \ell + q)!)^\beta \tau_{\ell,\beta,q}(\ell!)^\beta \left(\sum_{\substack{\mathbf{w} \leq \mathbf{v} \\ |\mathbf{w}|=\ell}} \binom{\mathbf{v}}{\mathbf{w}} \right)^\beta \\
&\leq (|\mathbf{v}|!)^\beta \sum_{\ell=0}^{|\mathbf{v}|-1} ((|\mathbf{v}| - \ell + q)!)^\beta \tau_{\ell,\beta,q} = (|\mathbf{v}|!)^\beta \tau_{|\mathbf{v}|,\beta,q},
\end{aligned}$$

which proves the statement. $\square$

We derive an explicit bound on $\tau_{|\mathbf{v}|,\beta,q}$.

Lemma 19. *There holds*

$$\tau_{k,\beta,q} \leq (q!)^{\beta k} 2^{\beta(q+1)k} \max\{1, 2^{k-1}\}.$$

Proof. We let $q \in \mathbb{Z}_+$ be arbitrary and prove the claim by induction with respect to $k \in \mathbb{Z}_+$. The basis of induction $k = 0$ follows from

$$\tau_{0,\beta,q} = 1.$$

Next, we fix $k \geq 1$ and assume that the claim is true for all values up to but not including k . Then, by utilizing the fact that $\frac{(k-\ell+q)!}{(k-\ell)!} = q! \binom{k-\ell+q}{q} \leq q! 2^{k-\ell+q}$, we obtain that

$$\begin{aligned}
\tau_{k,\beta,q} &= \sum_{\ell=0}^{k-1} \frac{((k-\ell+q)!)^\beta}{((k-\ell)!)^\beta} \tau_{\ell,\beta,q} \\
&\leq \sum_{\ell=0}^{k-1} (q!)^\beta 2^{\beta(k-\ell+q)} (q!)^\beta 2^{\beta(q+1)\ell} \max\{1, 2^{\ell-1}\} \\
&\leq (q!)^{\beta k} 2^{\beta k + \beta q + \beta q(k-1)} \sum_{\ell=0}^{k-1} \max\{1, 2^{\ell-1}\} \\
&= (q!)^{\beta k} 2^{\beta(q+1)k} 2^{k-1}
\end{aligned}$$

as desired. $\square$

Lemma 20. *Let $(\xi_{\mathbf{v}})_{\mathbf{v} \in \mathcal{F}}$ be a sequence satisfying the recurrence relation (4.3)–(4.4) with $C \geq 1$. Then, there holds*

$$\xi_{\mathbf{v}} \leq \sigma_{\min}^{-|\mathbf{v}|-1} C^{|\mathbf{v}|} (|\mathbf{v}|!)^\beta \max\{1, 2^{|\mathbf{v}|-1}\} \mathbf{b}^{\mathbf{v}} \text{ for all } \mathbf{v} \in \mathcal{F}.$$

Proof. We begin by proving the inequality

$$\xi_{\mathbf{v}} \leq \sigma_{\min}^{-|\mathbf{v}|-1} C^{|\mathbf{v}|} \alpha_{\mathbf{v}} \mathbf{b}^{\mathbf{v}} \text{ for all } \mathbf{v} \in \mathcal{F}, \quad (7.2)$$

where

$$\alpha_0 = 1 \quad \text{and} \quad \alpha_{\mathbf{v}} = \sum_{\mathbf{0} \neq \mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} (|\mathbf{m}|!)^\beta \alpha_{\mathbf{v}-\mathbf{m}}, \quad \mathbf{v} \in \mathcal{F} \setminus \{\mathbf{0}\}, \quad (7.3)$$

by induction over the multi-index $\mathbf{v}$.

The base case $\mathbf{v} = \mathbf{0}$ is given by

$$\xi_0 = \|\partial_{\mathbf{y}}^0 J(\cdot, \mathbf{y})^{-1}\|_{L^\infty} \leq \sigma_{\min}^{-1}.$$

For the induction hypothesis we fix a multi-index $\mathbf{v} \in \mathcal{F} \setminus \{\mathbf{0}\}$ and assume that the claim holds true for all multi-indices with order less than $|\mathbf{v}|$.

One finds

$$\begin{aligned} \xi_{\mathbf{v}} &\leq C \sigma_{\min}^{-1} \sum_{\mathbf{0} \neq \mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} (|\mathbf{m}|!)^\beta \mathbf{b}^{\mathbf{m}} \xi_{\mathbf{v}-\mathbf{m}} \\ &\leq C \sigma_{\min}^{-1} \sum_{\mathbf{0} \neq \mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} (|\mathbf{m}|!)^\beta \mathbf{b}^{\mathbf{m}} \sigma_{\min}^{-|\mathbf{v}|+|\mathbf{m}|-1} \max\{1, C\}^{|\mathbf{v}|-|\mathbf{m}|} \alpha_{\mathbf{v}-\mathbf{m}} \mathbf{b}^{\mathbf{v}-\mathbf{m}} \\ &\leq C^{|\mathbf{v}|} \sigma_{\min}^{-|\mathbf{v}|-1} \mathbf{b}^{\mathbf{v}} \sum_{\mathbf{0} \neq \mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} (|\mathbf{m}|!)^\beta \alpha_{\mathbf{v}-\mathbf{m}} \\ &= C^{|\mathbf{v}|} \sigma_{\min}^{-|\mathbf{v}|-1} \mathbf{b}^{\mathbf{v}} \alpha_{\mathbf{v}}. \end{aligned}$$

This proves inequality (7.2).

Next we show the following bound for the sequence (7.3):

$$\alpha_{\mathbf{v}} \leq \max\{1, 2^{|\mathbf{v}|-1}\} (|\mathbf{v}|!)^\beta.$$

We do this by induction. The induction base follows directly from the definition $\alpha_0 = 1$ and this in fact coincides with $\max\{1, 2^{0-1}\} (0!)^\beta = \max\{1, 2^{-1}\} = 1$.

Now let $\mathbf{v} \in \mathcal{F} \setminus \{\mathbf{0}\}$ and suppose that the claim has already been proven for all multi-indices with order less than $|\mathbf{v}|$. Then, applying the definition of the sequence α and the induction hypothesis, we obtain

$$\begin{aligned} \alpha_{\mathbf{v}} &\leq \sum_{\mathbf{0} \neq \mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} (|\mathbf{m}|!)^\beta \max\{1, 2^{|\mathbf{v}|-|\mathbf{m}|-1}\} ((|\mathbf{v}|-|\mathbf{m}|)!)^\beta \\ &= (|\mathbf{v}|!)^\beta + 2^{|\mathbf{v}|-1} \sum_{\mathbf{0} \neq \mathbf{m} < \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} (|\mathbf{m}|!)^\beta 2^{-|\mathbf{m}|} ((|\mathbf{v}|-|\mathbf{m}|)!)^\beta \\ &= (|\mathbf{v}|!)^\beta + 2^{|\mathbf{v}|-1} \sum_{\ell=1}^{|\mathbf{v}|-1} (\ell!)^\beta 2^{-\ell} ((|\mathbf{v}|-\ell)!)^\beta \frac{|\mathbf{v}|!}{\ell! (|\mathbf{v}|-\ell)!} \\ &\leq (|\mathbf{v}|!)^\beta + (|\mathbf{v}|!)^\beta 2^{|\mathbf{v}|-1} \underbrace{\sum_{\ell=1}^{|\mathbf{v}|-1} 2^{-\ell}}_{=1-2^{1-|\mathbf{v}|}} \end{aligned}$$

$$= (|\mathbf{v}|!)^\beta 2^{|\mathbf{v}|-1}.$$

Note that this is precisely $\alpha_{\mathbf{v}}$ when $\mathbf{v} \neq \mathbf{0}$, which completes the induction proof. $\square$

The following result is used to resolve the inductive bounds for both the stationary and parabolic settings.

Lemma 21. *Let $(\Lambda_{\mathbf{v}})_{\mathbf{v} \in \mathcal{F}}$ satisfy, for all $\mathbf{v} \in \mathcal{F} \setminus \{\mathbf{0}\}$,*

$$\Lambda_{\mathbf{0}} \leq C_0,$$

$$\Lambda_{\mathbf{v}} \leq C \sum_{\substack{\mathbf{m} \leq \mathbf{v} \\ \mathbf{m} \neq \mathbf{0}}} \binom{\mathbf{v}}{\mathbf{m}} ((|\mathbf{m}| + k)!)^\beta (C\mathbf{b})^{\mathbf{m}} \Lambda_{\mathbf{v}-\mathbf{m}} + C((|\mathbf{v}| + k)!)^\beta (C\mathbf{b})^{\mathbf{v}},$$

where $C_0, C > 0$ are constants and $k \in \mathbb{N}$. Then there holds

$$\Lambda_{\mathbf{v}} \leq (1 + C_0)(C\mathbf{b})^{\mathbf{v}} (|\mathbf{v}|!)^\beta \tilde{\tau}_{|\mathbf{v}|, \beta, k},$$

where

$$\tilde{\tau}_{0, \beta, k} = 1 \quad \text{and} \quad \tilde{\tau}_{\mathbf{v}, \beta, k} = C \sum_{\ell=0}^{v-1} \frac{((v-\ell+k)!)^\beta}{((v-\ell)!)^\beta} \tilde{\tau}_{\ell, \beta, k}. \quad (7.4)$$

Proof. By the induction hypothesis, we obtain

$$\begin{aligned} \Lambda_{\mathbf{v}} &\leq C \sum_{\substack{\mathbf{m} \leq \mathbf{v} \\ \mathbf{m} \neq \mathbf{0} \\ \mathbf{m} \neq \mathbf{v}}} \binom{\mathbf{v}}{\mathbf{m}} ((|\mathbf{m}| + k)!)^\beta (C\mathbf{b})^{\mathbf{m}} (1 + C_0)(C\mathbf{b})^{\mathbf{v}-\mathbf{m}} (|\mathbf{v}|-|\mathbf{m}|!)^\beta \tilde{\tau}_{|\mathbf{v}|-|\mathbf{m}|, \beta, k} \\ &\quad + (C_0 + 1)C((|\mathbf{v}| + k)!)^\beta (C\mathbf{b})^{\mathbf{v}} \\ &= (C_0 + 1)C(C\mathbf{b})^{\mathbf{v}} \sum_{\ell=1}^{|\mathbf{v}|-1} ((\ell + k)!)^\beta ((|\mathbf{v}| - \ell)!)^\beta \tilde{\tau}_{|\mathbf{v}|-\ell, \beta, k} \sum_{\substack{|\mathbf{m}|=\ell \\ \mathbf{m} \leq \mathbf{v}}} \binom{\mathbf{v}}{\mathbf{m}} \\ &\quad + (C_0 + 1)C((|\mathbf{v}| + k)!)^\beta (C\mathbf{b})^{\mathbf{v}} \\ &= (C_0 + 1)C(C\mathbf{b})^{\mathbf{v}} \sum_{\ell=1}^{|\mathbf{v}|} ((\ell + k)!)^\beta ((|\mathbf{v}| - \ell)!)^\beta \tilde{\tau}_{|\mathbf{v}|-\ell, \beta, k} \sum_{\substack{|\mathbf{m}|=\ell \\ \mathbf{m} \leq \mathbf{v}}} \binom{\mathbf{v}}{\mathbf{m}} \\ &\leq (C_0 + 1)C(C\mathbf{b})^{\mathbf{v}} (|\mathbf{v}|!)^\beta \sum_{\ell=1}^{|\mathbf{v}|} \frac{((\ell + k)!)^\beta}{(\ell!)^\beta} \tilde{\tau}_{|\mathbf{v}|-\ell, \beta, k} \\ &= (C_0 + 1)C(C\mathbf{b})^{\mathbf{v}} (|\mathbf{v}|!)^\beta \sum_{\ell=0}^{|\mathbf{v}|-1} \frac{((|\mathbf{v}| - \ell + k)!)^\beta}{(|\mathbf{v}| - \ell!)^\beta} \tilde{\tau}_{\ell, \beta, k} \\ &= (C_0 + 1)(C\mathbf{b})^{\mathbf{v}} (|\mathbf{v}|!)^\beta \tilde{\tau}_{|\mathbf{v}|, \beta, k}, \end{aligned}$$

as desired. $\square$

Lemma 22. *Under Assumptions (A1)–(A2), (A4) as well as (A9) the solution $\hat{u}$ of the variational problem*

$$b(\mathbf{y}; \hat{u}, \hat{v}) = F(\mathbf{y}; \hat{v}) \quad \text{for all } \hat{v} = (\hat{v}_1, \hat{v}_2) \in \mathcal{V} \text{ and } \mathbf{y} \in U,$$

where

$$\begin{aligned} b(\mathbf{y}; \hat{u}, \hat{v}) &:= \int_I \left\langle \frac{\partial}{\partial t} \hat{u}(\cdot, t, \mathbf{y}), \hat{v}_1(\cdot, t) \det J(\cdot, \mathbf{y}) \right\rangle_{H^{-1}, H_0^1} dt \\ &\quad + \int_I \int_{D_{\text{ref}}} A(\mathbf{x}, \mathbf{y}) \nabla \hat{u}(\mathbf{x}, t, \mathbf{y}) \cdot \nabla \hat{v}_1(\mathbf{x}, t) d\mathbf{x} dt \\ &\quad + \int_{D_{\text{ref}}} \hat{u}(\mathbf{x}, 0, \mathbf{y}) \hat{v}_2(\mathbf{x}) \det J(\mathbf{x}, \mathbf{y}) d\mathbf{x} \end{aligned}$$

and

$$\begin{aligned} F(\mathbf{y}; \hat{v}) &:= \int_I \int_{D_{\text{ref}}} f_{\text{ref}}(\mathbf{x}, \mathbf{y}) \hat{v}_1(\mathbf{x}, t) d\mathbf{x} dt \\ &\quad + \int_{D_{\text{ref}}} \hat{u}_0(\mathbf{x}, \mathbf{y}) \hat{v}_2(\mathbf{x}) \det J(\mathbf{x}, \mathbf{y}) d\mathbf{x} \end{aligned}$$

has the property

$$\partial_{\mathbf{y}}^{\mathbf{v}} \hat{u}(\cdot, \cdot, \mathbf{y}) \in L^2(I; H_0^1(D_{\text{ref}}))$$

for all $\mathbf{v} \in \mathcal{F}$.

Proof. We will perform a proof by induction over the differentiation order $\mathbf{v}$ and show that the parametric derivatives lie in the space $\mathcal{X}$.

For $\mathbf{v} = \mathbf{0}$ we have that $\hat{u}(\cdot, \cdot, \mathbf{y}) \in \mathcal{X}$, which holds true. We fix $\mathbf{v} \in \mathcal{F} \setminus \{\mathbf{0}\}$ and assume that the claim holds true for all multi-indices with order less than $|\mathbf{v}|$, which means that all the lower order parametric derivatives of $\hat{u}$ lie in $\mathcal{X}$.

In order to find the correct operator equation later used in the proof, we formally differentiate the equation

$$b(\mathbf{y}; \hat{u}, \hat{v}) = F(\mathbf{y}; \hat{v})$$

with respect to $\mathbf{y} \in U$, where $\hat{v} = (\hat{v}_1, \hat{v}_2) \in \mathcal{V}$, i.e.,

$$\partial_{\mathbf{y}}^{\mathbf{v}} b(\mathbf{y}; \hat{u}, \hat{v}) = \partial_{\mathbf{y}}^{\mathbf{v}} F(\mathbf{y}; \hat{v}).$$

By Leibniz product rule we get

$$\sum_{\mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} \left[\int_I \left\langle \frac{\partial}{\partial t} (\partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \hat{u}(\cdot, t, \mathbf{y})), \hat{v}_1(\cdot, t) \partial_{\mathbf{y}}^{\mathbf{m}} \det J(\cdot, \mathbf{y}) \right\rangle_{H^{-1}, H_0^1} dt \right.$$

$$\begin{aligned}
& + \int_I \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{m}} A(\mathbf{x}, \mathbf{y}) \nabla \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \hat{u}(\cdot, t, \mathbf{y}) \cdot \nabla \hat{v}_1(\mathbf{x}, t) \, d\mathbf{x} \, dt \\
& + \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \hat{u}(\mathbf{x}, 0, \mathbf{y}) \hat{v}_2(\mathbf{x}) \partial_{\mathbf{y}}^{\mathbf{m}} \det J(\mathbf{x}, \mathbf{y}) \, d\mathbf{x} \Big] \\
& = \int_I \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{v}} f_{\text{ref}}(\cdot, \mathbf{y}) \hat{v}_1 \, d\mathbf{x} \, dt \\
& + \sum_{\mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \hat{u}_0(\mathbf{x}, \mathbf{y}) \hat{v}_2(\mathbf{x}) \partial_{\mathbf{y}}^{\mathbf{m}} \det J(\mathbf{x}, \mathbf{y}) \, d\mathbf{x},
\end{aligned}$$

where $\hat{u}_0(\mathbf{x}, \mathbf{y}) = u_0(\mathbf{V}(\mathbf{x}, \mathbf{y}), \mathbf{y})$ for $\mathbf{x} \in D_{\text{ref}}, \mathbf{y} \in U$.

Isolating the case $\mathbf{m} = \mathbf{0}$ we can rewrite this as

$$\begin{aligned}
& \int_I \langle \frac{\partial}{\partial t} (\partial_{\mathbf{y}}^{\mathbf{v}} \hat{u}(\cdot, t, \mathbf{y})), \hat{v}_1(\cdot, t) \det J(\cdot, \mathbf{y}) \rangle_{H^{-1}, H_0^1} \, dt \\
& + \int_I \int_{D_{\text{ref}}} A(\mathbf{x}, \mathbf{y}) \nabla \partial_{\mathbf{y}}^{\mathbf{v}} \hat{u}(\mathbf{x}, t, \mathbf{y}) \cdot \nabla \hat{v}_1(\mathbf{x}, t) \, d\mathbf{x} \, dt \\
& + \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{v}} \hat{u}(\mathbf{x}, 0, \mathbf{y}) \hat{v}_2(\mathbf{x}) \det J(\mathbf{x}, \mathbf{y}) \, d\mathbf{x} \\
& + \sum_{\mathbf{0} \neq \mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} \left[\int_I \langle \frac{\partial}{\partial t} (\partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \hat{u}(\cdot, t, \mathbf{y})), \hat{v}_1(\cdot, t) \partial_{\mathbf{y}}^{\mathbf{m}} \det J(\cdot, \mathbf{y}) \rangle_{H^{-1}, H_0^1} \, dt \right. \\
& + \int_I \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{m}} A(\mathbf{x}, \mathbf{y}) \nabla \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \hat{u}(\mathbf{x}, t, \mathbf{y}) \cdot \nabla \hat{v}_1(\mathbf{x}, t) \, d\mathbf{x} \, dt \\
& \left. + \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \hat{u}(\mathbf{x}, 0, \mathbf{y}) \hat{v}_2(\mathbf{x}) \partial_{\mathbf{y}}^{\mathbf{m}} \det J(\mathbf{x}, \mathbf{y}) \, d\mathbf{x} \right] \\
& = \int_I \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{v}} f_{\text{ref}}(\mathbf{x}, \mathbf{y}) \hat{v}_1(\mathbf{x}) \, d\mathbf{x} \, dt \\
& + \sum_{\mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \hat{u}_0(\mathbf{x}, \mathbf{y}) \hat{v}_2(\mathbf{x}) \partial_{\mathbf{y}}^{\mathbf{m}} \det J(\mathbf{x}, \mathbf{y}) \, d\mathbf{x},
\end{aligned}$$

Denoting

$$\begin{aligned}
T_{\mathbf{m}} & := \int_I \langle \frac{\partial}{\partial t} (\partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \hat{u}(\cdot, t, \mathbf{y})), \hat{v}_1(\cdot, t) \partial_{\mathbf{y}}^{\mathbf{m}} \det J(\cdot, \mathbf{y}) \rangle_{H^{-1}, H_0^1} \, dt \\
& + \int_I \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{m}} A(\mathbf{x}, \mathbf{y}) \nabla \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \hat{u}(\mathbf{x}, t, \mathbf{y}) \cdot \nabla \hat{v}_1(\mathbf{x}, t) \, d\mathbf{x} \, dt \\
& + \int_{D_{\text{ref}}} \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \hat{u}(\mathbf{x}, 0, \mathbf{y}) \hat{v}_2(\mathbf{x}) \partial_{\mathbf{y}}^{\mathbf{m}} \det J(\mathbf{x}, \mathbf{y}) \, d\mathbf{x}
\end{aligned}$$

we can write

$$b(\mathbf{y}; \partial_{\mathbf{y}}^{\mathbf{v}} \hat{u}, \hat{v}) = \partial_{\mathbf{y}}^{\mathbf{v}} F(\mathbf{y}; \hat{v}) - \sum_{\mathbf{0} \neq \mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} T_{\mathbf{m}}(\mathbf{y}; \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \hat{u}, \hat{v}). \quad (7.5)$$

Note that these terms are well-defined due to the induction hypothesis.

Now define a functional $R_{\mathbf{v}}(\mathbf{y}) \in \mathcal{Y}^*$ by

$$\langle R_{\mathbf{v}}(\mathbf{y}), \widehat{v} \rangle := \partial_{\mathbf{y}}^{\mathbf{v}} F(\mathbf{y}; \widehat{v}) - \sum_{\mathbf{0} \neq \mathbf{m} \leq \mathbf{v}} \binom{\mathbf{v}}{\mathbf{m}} T_{\mathbf{m}}(\mathbf{y}; \partial_{\mathbf{y}}^{\mathbf{v}-\mathbf{m}} \widehat{u}, \widehat{v})$$

as well as $\mathcal{B}(\mathbf{y}) : \mathcal{X} \rightarrow \mathcal{Y}^*$ by

$$\langle \mathcal{B}(\mathbf{y})w, \widehat{v} \rangle = b(\mathbf{y}; w, \widehat{v}), \quad w \in \mathcal{X}.$$

Then, the above identity (7.5) can be expressed as

$$\mathcal{B}(\mathbf{y})w = R_{\mathbf{v}}(\mathbf{y}).$$

For this operator equation to make sense we note that for $w \in \mathcal{X}$ and $\widehat{v} \in \mathcal{Y}$ there holds

$$\begin{aligned} |T_{\mathbf{m}}(\mathbf{y}; w, \widehat{v})| &\leq \int_I \left| \left\langle \frac{\partial}{\partial t} w \partial_{\mathbf{y}}^{\mathbf{m}} \det J(\cdot, \mathbf{y}), \widehat{v}_1(\cdot, t) \right\rangle_{H^{-1}, H_0^1} \right| dt \\ &\quad + \int_I \int_{D_{\text{ref}}} |\partial_{\mathbf{y}}^{\mathbf{m}} A(\mathbf{x}, \mathbf{y}) \nabla w \cdot \nabla \widehat{v}_1(\mathbf{x}, t)| \, d\mathbf{x} \, dt \\ &\quad + \int_{D_{\text{ref}}} |w(\cdot, 0) \widehat{v}_2(\mathbf{x}) \partial_{\mathbf{y}}^{\mathbf{m}} \det J(\mathbf{x}, \mathbf{y})| \, d\mathbf{x} \\ &\leq C \|\partial_{\mathbf{y}}^{\mathbf{m}} \det J(\cdot, \mathbf{y})\|_{W^{1,\infty}} \left\| \frac{\partial}{\partial t} w \right\|_{L^2(I; H^{-1})} \|\widehat{v}_1\|_{L^2(I; H_0^1)} \\ &\quad + \|\partial_{\mathbf{y}}^{\mathbf{m}} A(\cdot, \mathbf{y})\|_{L^\infty} \int_I \int_{D_{\text{ref}}} |\nabla w \cdot \nabla \widehat{v}_1(\mathbf{x}, t)| \, d\mathbf{x} \, dt \\ &\quad + \|\partial_{\mathbf{y}}^{\mathbf{m}} \det J(\cdot, \mathbf{y})\|_{L^\infty} \int_{D_{\text{ref}}} |w(\cdot, 0) \widehat{v}_2(\mathbf{x})| \, d\mathbf{x}. \end{aligned} \tag{7.6}$$

This observation and applying Cauchy–Schwarz inequality to the other two summands leads to a bound of the expression (7.6) given by

$$\begin{aligned} &C \|\partial_{\mathbf{y}}^{\mathbf{m}} \det J(\cdot, \mathbf{y})\|_{W^{1,\infty}} \left\| \frac{\partial}{\partial t} w \right\|_{L^2(I; H^{-1})} \|\widehat{v}_1\|_{L^2(I; H_0^1)} \\ &\quad + \|\partial_{\mathbf{y}}^{\mathbf{m}} A(\cdot, \mathbf{y})\|_{L^\infty} \|\nabla w\|_{L^2(D \times I)} \|\nabla \widehat{v}_1\|_{L^2(D \times I)} \\ &\quad + \|\partial_{\mathbf{y}}^{\mathbf{m}} \det J(\cdot, \mathbf{y})\|_{L^\infty} \|w(0)\|_{L^2} \|\widehat{v}_2\|_{L^2} \\ &= C \|\partial_{\mathbf{y}}^{\mathbf{m}} \det J(\cdot, \mathbf{y})\|_{W^{1,\infty}} \\ &\quad \times (\left\| \frac{\partial}{\partial t} w \right\|_{L^2(I; H^{-1})} \|\widehat{v}_1\|_{L^2(I; H_0^1)} + \|w(0)\|_{L^2} \|\widehat{v}_2\|_{L^2}) \\ &\quad + \|\partial_{\mathbf{y}}^{\mathbf{m}} A(\cdot, \mathbf{y})\|_{L^\infty} \|\nabla w\|_{L^2(D \times I)} \|\nabla \widehat{v}_1\|_{L^2(D \times I)} \\ &\leq (1 + C_{\text{tr}}) C \|\partial_{\mathbf{y}}^{\mathbf{m}} \det J(\cdot, \mathbf{y})\|_{W^{1,\infty}} \|w\|_{\mathcal{X}} \|\widehat{v}\|_{\mathcal{Y}} \\ &\quad + \|\partial_{\mathbf{y}}^{\mathbf{m}} A(\cdot, \mathbf{y})\|_{L^\infty} \|w\|_{\mathcal{X}} \|\widehat{v}\|_{\mathcal{Y}} \\ &\leq C(\mathbf{y}, \mathbf{m}) \|w\|_{\mathcal{X}} \|\widehat{v}\|_{\mathcal{Y}}, \end{aligned}$$

where $C(\mathbf{y}, \mathbf{m}) := (1 + C_{\text{tr}})C\|\partial_{\mathbf{y}}^{\mathbf{m}} \det J(\cdot, \mathbf{y})\|_{W^{1,\infty}} + \|\partial_{\mathbf{y}}^{\mathbf{m}} A(\cdot, \mathbf{y})\|_{L^\infty}$. Note that in the penultimate inequality we use the fact that

$$\mathcal{X} \hookrightarrow \mathcal{C}([0, T], L^2(D_{\text{ref}}))$$

and hence $\|w(0)\|_{L^2} \leq C_{\text{tr}}\|w\|_{\mathcal{X}}$ (cf. Evans (2010, Theorem 5.9.3)). It follows that $T_{\mathbf{m}} \in \mathcal{Y}^*$. Exploiting the bounds found in the regularity analysis, one easily sees that a similar reasoning reveals that $\partial_{\mathbf{y}}^{\mathbf{v}} F(\mathbf{y}; \cdot) \in \mathcal{Y}^*$. Thus, $R_{\mathbf{v}}(\mathbf{y}) \in \mathcal{Y}^*$.

By Assumption (A2), the Jacobian determinant $\det J(\cdot, \mathbf{y})$ is bounded above and below away from zero and the coefficient matrix

$$A(\mathbf{x}, \mathbf{y}) = (J(\mathbf{x}, \mathbf{y})^T J(\mathbf{x}, \mathbf{y}))^{-1} \det J(\mathbf{x}, \mathbf{y})$$

is uniformly bounded as well as uniformly elliptic. In consequence, the pullback space-time bilinear form

$$\begin{aligned} b(\mathbf{y}; w, \widehat{v}) &= \int_I \langle \partial_t w(t), \widehat{v}_1(t) \rangle_{H^{-1}, H_0^1} dt \\ &\quad + \int_I \int_{D_{\text{ref}}} A(\mathbf{x}, \mathbf{y}) \nabla w \cdot \nabla \widehat{v}_1 \, d\mathbf{x} dt \\ &\quad + \int_{D_{\text{ref}}} w(\mathbf{x}, 0) \widehat{v}_2(\mathbf{x}) \det J(\mathbf{x}, \mathbf{y}) \, d\mathbf{x} \end{aligned}$$

satisfies the boundedness and coercivity assumptions of Schwab and Stevenson (2009, Theorem 5.1). Therefore, the induced operator

$$\mathcal{B}(\mathbf{y}) : \mathcal{X} \rightarrow \mathcal{Y}^*, \quad \langle \mathcal{B}(\mathbf{y})w, \widehat{v} \rangle := b(\mathbf{y}; w, \widehat{v}),$$

is boundedly invertible.

By the bounded invertibility of $\mathcal{B}(\mathbf{y})$ it follows that there exists a unique solution $w_{\mathbf{v}} \in \mathcal{X}$ of the equation $\mathcal{B}(\mathbf{y})w_{\mathbf{v}} = R_{\mathbf{v}}(\mathbf{y})$. By uniqueness and the fact that we have seen that the operator equation is fulfilled by $\partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\cdot, \cdot, \mathbf{y})$ it follows that $w_{\mathbf{v}} = \partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\cdot, \cdot, \mathbf{y})$. Hence, $\partial_{\mathbf{y}}^{\mathbf{v}} \widehat{u}(\cdot, \cdot, \mathbf{y}) \in \mathcal{X}$. Since $\mathcal{X} \subset L^2(I; H_0^1(D_{\text{ref}}))$ the statement is verified. $\square$

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